

Study Statistics

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Introduction

This document details the R code used in the analysis of heart rate and demographic data for a cohort presenting to triage in Masindi, Uganda. Some basic descriptive statistics were performed elsewhere and are therefore not covered here.

Additionally, this analysis includes a comparison of the Masindi cohort with frequently cited data from the United States, sourced from the NHANES (National Health and Nutrition Examination Survey) datasets. These datasets are publicly available at <https://wwwn.cdc.gov/nchs/nhanes/default.aspx> and were accessed in January 2025. Data from the years 1999–2008 were selected to align with the influential NHSR heart rate publication:

Ostchega, Y., Porter, K. S., Hughes, J., Dillon, C. F., & Nwankwo, T. (2011). Resting pulse rate reference data for children, adolescents, and adults; United States, 1999-2008.

File Structure

All files associated with this analysis are organized within the `main` folder:

1. Raw Data:

- NHANES raw data is located in `main\Raw Data\NHANES`.
- Masindi raw data is located in `main\Raw Data\Masindi`.

2. Code: The `main\Code` folder contains all R scripts used in the analysis:

- `Masindi_Data_Prep.R`
- `NHANES_Data_Prep.R`
- `Masindi_HR_Analysis.R`
- `Masindi_Census_Comparison.R`
- `NHANES_Masindi_Comparison.R`

Workflow Summary

1. Data Preparation:

- `Masindi_Data_Prep.R` and `NHANES_Data_Prep.R` transform raw data into ready-to-use datasets:
 - `Masindi_data.csv`
 - `Masindi_routine_data.csv`
 - `HR_DEMO_NHANES.csv`
- These processed files, as well as the Masindi district 2022 census projections (`Masindi_Census_Projections.csv`), are saved in `main\Processed Data`.

2. Statistical Analyses:

- `Masindi_HR_Analysis.R` performs statistical analyses within the Masindi cohort.
- `Masindi_Census_Comparison.R` extrapolates Masindi cohort data to the district level.
- `NHANES_Masindi_Comparison.R` compares heart rate and demographic data between the Masindi and NHANES cohorts.

File Name	Path	Description
Masindi Data.csv	Main\Raw Data\Masindi	Raw health (including heart rates) and demographic data for the Masindi cohort.
BPX_1999_2000.xpt BPX_2001_2002.xpt BPX_2003_2004.xpt BPX_2005_2006.xpt BPX_2007_2008.xpt	Main\Raw Data\NHANES	Blood pressure examination results (including heart rates) for the NHANES cohort for two given years.
DEMO_1999_2000.xpt DEMO_2001_2002.xpt DEMO_2003_2004.xpt DEMO_2005_2006.xpt DEMO_2007_2008.xpt	Main\Raw Data\NHANES	Corresponding demographic data for the NHANES cohort for two given years.
Masindi_Data_Prep.R	Main\Code	R script used for the cleaning and formatting of the Masindi data. This resulted in two files used in analysis: the Masindi_data.csv file with all presenting patients and the Masindi_routine_data.csv file, which includes routine clinic visits but not those presenting with new concerns.
NHANES_Data_Prep.R	Main\Code	R script used for the merging, cleaning, and formatting of the NHANES data. This resulted in the HR_DEMO_NHANES.csv file, containing both participant heart rate and demographic data for analysis.
Masindi_data.csv	Main\Processed Data\Masindi	Prepared dataset used for analysis.
Masindi_routine_data.csv	Main\Processed Data\Masindi	Same dataset as Masindi_data.csv, except only those presenting for routine visits are included.
Masindi_Census_Projections.csv	Main\Processed Data\Masindi	The projected 2022 demographic structure for Masindi district was distilled from the census available at: https://www.masindi.go.ug/district/population
HR_DEMO_NHANES.csv	Main\Processed Data\NHANES	Prepared dataset used for analysis.
Masindi_HR_Analysis.R	Main\Code	R script used to assess determinants of heart rate within the Masindi group. Includes both analysis including and excluding those presenting with new concerns.
Masindi_Census_Comparison.R	Main\Code	R script used for formatting Masindi district census data and projecting cohort heart rates to district-level demographic structure.
NHANES_Masindi_Comparison.R	Main\Code	R script used to compare heart rates between the Masindi and NHANES cohorts, while adjusting for various covariates.

Data Preparation

Patient data collected from Masindi were cleaned and formatted using the R script `Masindi_Data_Prep.R`. This script was used to create two datasets: `Masindi_data.csv` and `Masindi_routine_data.csv`. The former includes all presenting patients, while the latter includes only those presenting for routine visits. The NHANES data was cleaned and formatted using the R script `NHANES_Data_Prep.R`. This script was used to create the dataset `HR_DEMO_NHANES.csv`, which includes both participant heart rate and demographic data for analysis.

Masindi_Data_Prep.R

```
# Load necessary libraries
library(tidyverse)
library(Hmisc)
library(knitr)

folder_path <- as.character(here::here()) #>>>THIS SHOULD POINT TO THE 'MAIN' FOLDER<<<#

# Read Heart Rate Data
HR_data <- read.csv(paste0(folder_path, "/Raw Data/Masindi/Masindi Data.csv"))

# Setting Labels and Factors
label(HR_data$record_id) = "Record ID"
label(HR_data$age) = "Age"
label(HR_data$gender) = "Gender"
label(HR_data$reason) = "Reason for visit"
label(HR_data$hr) = "x6.lead.Heart.Rate"

HR_data$redcap_repeat_instrument.factor = factor(HR_data$redcap_repeat_instrument,
  levels = c("form_1", "ekg_interpretation"))
```

Rows missing demographic forms or heart rate measurements are removed. Selected columns include the patient ID (`Masindi_ID`), heart rate (beats per minute, `hr`), age (years, `age`), sex (`sex`), and reason for visit (`reason`).

```
# Filter and clean up data
HR_data <- HR_data %>%
  filter(redcap_repeat_instrument.factor == "form_1") %>%
  mutate(hr = X6.lead.Heart.Rate) %>%
  filter(!is.na(hr))

HR_data_export <- HR_data %>%
  select(Masindi_ID = record_id, hr = hr, age = age, sex = gender,
    reason = reason) %>%
  mutate(source = "Masindi", sex = case_when(sex == 1 ~ "F",
    sex == 2 ~ "M", TRUE ~ as.factor(sex)), reason = case_when(reason ==
    1 ~ "New Concern", reason != 1 ~ "Routine", TRUE ~ as.factor(reason)))
```

Patients presenting with new concerns are identified by `reason = 1` and are removed from the routine only data set.


```
HR_data_export_routineonly <- HR_data %>%
  filter(reason != 1) %>%
  select(Masindi_ID = record_id, hr = hr, age = age, sex = gender) %>%
  mutate(source = "Masindi", sex = case_when(sex == 1 ~ "F",
    sex == 2 ~ "M", TRUE ~ as.factor(sex)))

any(is.na(HR_data_export)) # No NAs
```

```
## [1] FALSE
```

```
any(is.na(HR_data_export_routineonly)) # No NAs
```

```
## [1] FALSE
```

Finally, the two prepared dataframes are exported to be used in analysis.

```
# Create files for processed data
write_csv(HR_data_export, paste0(folder_path, "/Processed Data/Masindi/Masindi_data.csv"))
write_csv(HR_data_export_routineonly, paste0(folder_path, "/Processed Data/Masindi/Masindi_routine_data.csv"))
```

NHANES_Data_Prep.R

```
library(tidyverse)
library(haven)
```

```
folder_path <- as.character(here::here()) #>>>THIS SHOULD POINT TO THE 'MAIN' FOLDER<<<#

# Data Import
BPX_1999_2000 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/BPX_1999_2000.xpt"))
BPX_2001_2002 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/BPX_2001_2002.xpt"))
BPX_2003_2004 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/BPX_2003_2004.xpt"))
BPX_2005_2006 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/BPX_2005_2006.xpt"))
BPX_2007_2008 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/BPX_2007_2008.xpt"))
DEMO_1999_2000 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/DEMO_1999_2000.xpt"))
DEMO_2001_2002 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/DEMO_2001_2002.xpt"))
DEMO_2003_2004 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/DEMO_2003_2004.xpt"))
DEMO_2005_2006 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/DEMO_2005_2006.xpt"))
DEMO_2007_2008 <- read_xpt(paste0(folder_path, "/Raw Data/NHANES/DEMO_2007_2008.xpt"))
```

After importing the NHANES blood pressure examination (BPX) and demographic (DEMO) datasheets, the dataframes were merged using the "SEQN" column, which represents a unique ID for each participant across years.

```
HR_DEMO_1999_2000 <- BPX_1999_2000 %>%
  select(SEQN, BPXP1S) %>%
  inner_join(DEMO_1999_2000, by = "SEQN") %>%
  select(1:7)

HR_DEMO_2001_2002 <- BPX_2001_2002 %>%
```

```

select(SEQN, BPXPLS) %>%
inner_join(DEMO_2001_2002, by = "SEQN") %>%
select(1:7)

HR_DEMO_2003_2004 <- BPX_2003_2004 %>%
select(SEQN, BPXPLS) %>%
inner_join(DEMO_2003_2004, by = "SEQN") %>%
select(1:7)

HR_DEMO_2005_2006 <- BPX_2005_2006 %>%
select(SEQN, BPXPLS) %>%
inner_join(DEMO_2005_2006, by = "SEQN") %>%
select(1:7)

HR_DEMO_2007_2008 <- BPX_2007_2008 %>%
select(SEQN, BPXPLS) %>%
inner_join(DEMO_2007_2008, by = "SEQN") %>%
select(1:7)

```

The dataframes from different years were then combined into a single one. The data was filtered to include only adults (age 18 and older) and those with pulse data. The columns were selected and renamed to match the Masindi dataset.

```

HR_DEMO <- bind_rows(HR_DEMO_1999_2000, HR_DEMO_2001_2002, HR_DEMO_2003_2004, HR_DEMO_2005_2006, HR_DEMO_2007_2008)
any(duplicated(HR_DEMO$SEQN)) # FALSE --> NO repeated SEQN (Identification Codes)

```

```
## [1] FALSE
```

```

HR_DEMO <- HR_DEMO %>%
  filter(RIDAGEYR >= 18) %>% # Adults only
  filter(!is.na(BPXPLS)) %>% # Must have Pulse data
  select(NHANES_ID = SEQN, hr = BPXPLS, age = RIDAGEYR, sex = RIAGENDR) %>% # 1 = MALE, 2 = FEMALE
  mutate(source = "NHANES",
    sex = case_when(
      sex == 1 ~ "M",
      sex == 2 ~ "F",
      TRUE ~ as.factor(sex)
    ))

write_csv(HR_DEMO, paste0(folder_path, "/Processed Data/NHANES/HR_DEMO_NHANES.csv"))

```

Statistical Analysis

Data analysis involved three steps: assessing the determinants of heart rates within Masindi, evaluating demographic differences between Masindi and NHANES, and a comparison of heart rates between the two groups. The first two steps were crucial in identifying what parameters could be adjusted for when making the final comparison. For the Masindi dataset, all statistics were repeated with the exclusion of “new concern” patients. The R scripts `Masindi_HR_Analysis.R` and `NHANES_Masindi_Comparison.R` were used for these analyses.

Masindi_HR_Analysis.R

All Masindi

```
# Load necessary libraries
library(tidyverse)
library(lme4)
library(patchwork)
library(ggpubr)
library(knitr)
```

```
folder_path <- as.character(here::here()) #>>>THIS SHOULD POINT TO THE 'MAIN' FOLDER<<<#
```

```
# Data Import
Masindi <- read_csv(paste0(folder_path, "/Processed Data/Masindi/Masindi_data.csv"))
```

```
## Rows: 1059 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (3): sex, reason, source
## dbl (3): Masindi_ID, hr, age
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
Masindi_routine <- read_csv(paste0(folder_path, "/Processed Data/Masindi/Masindi_routine_data.csv"))
```

```
## Rows: 388 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (2): sex, source
## dbl (3): Masindi_ID, hr, age
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

After loading the appropriate libraries and importing data, a publication-ready figure was produced, showing the difference of mean heart rates by sex and by reason for visit.

```

### Figure

categories <- factor(c("Male", "Female", "Routine", "New Concern"),
  levels = c("Male", "Female", "Routine", "New Concern"))
means <- c(mean(Masindi$hr[Masindi$sex == "M"]), mean(Masindi$hr[Masindi$sex ==
  "F"]), mean(Masindi$hr[Masindi$reason == "Routine"]), mean(Masindi$hr[Masindi$reason ==
  "New Concern"]))

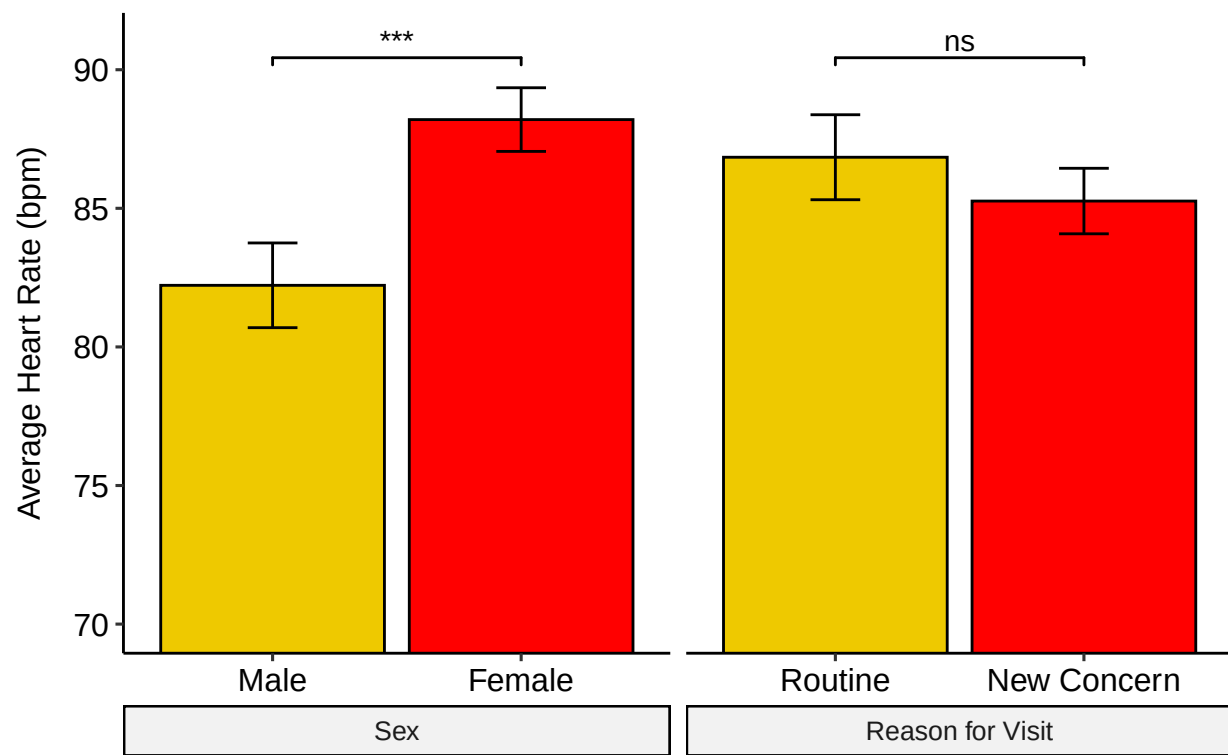
ci_lower <- c(t.test(Masindi$hr[Masindi$sex == "M"])$conf.int[1],
  t.test(Masindi$hr[Masindi$sex == "F"])$conf.int[1], t.test(Masindi$hr[Masindi$reason ==
  "Routine"])$conf.int[1], t.test(Masindi$hr[Masindi$reason ==
  "New Concern"])$conf.int[1])

ci_upper <- c(t.test(Masindi$hr[Masindi$sex == "M"])$conf.int[2],
  t.test(Masindi$hr[Masindi$sex == "F"])$conf.int[2], t.test(Masindi$hr[Masindi$reason ==
  "Routine"])$conf.int[2], t.test(Masindi$hr[Masindi$reason ==
  "New Concern"])$conf.int[2])

plot_data <- data.frame(Category = categories, Mean = means,
  Lower = ci_lower, Upper = ci_upper)
plot_data$FillColor <- c("gold2", "red", "gold2", "red") # Assign colors manually
plot_data$wrapping <- factor(c("Sex", "Sex", "Reason", "Reason"),
  levels = c("Sex", "Reason"))

# tiff('FigA.tiff', units='px', width=1900, height=2300,
# res=400)
ggplot(plot_data, aes(x = factor(Category, levels = unique(Category[wrapping ==
  wrapping])), y = Mean, fill = FillColor)) + geom_bar(stat = "identity",
  color = "black") + geom_errorbar(aes(ymin = Lower, ymax = Upper),
  width = 0.2) + scale_y_continuous(breaks = seq(70, 91, by = 5)) +
  coord_cartesian(ylim = c(70, 91)) + scale_fill_identity() +
  labs(x = "", y = "Average Heart Rate (bpm)", title = "") +
  facet_grid(cols = vars(wrapping), scales = "free_x", space = "free_x",
  switch = "x", labeller = labeller(wrapping = c(Reason = "Reason for Visit"))) +
  theme_pubr() + geom_signif(comparisons = list(c("Male", "Female")),
  annotations = "****", y_position = 90, tip_length = 0.03,
  data = plot_data[plot_data$wrapping == "Sex", ]) + geom_signif(comparisons = list(c("Routine",
  "New Concern")), annotations = "ns", y_position = 90, tip_length = 0.03,
  data = plot_data[plot_data$wrapping == "Reason", ]) + theme(strip.placement = "outside",
  axis.title.y = element_text(margin = margin(r = 10)))

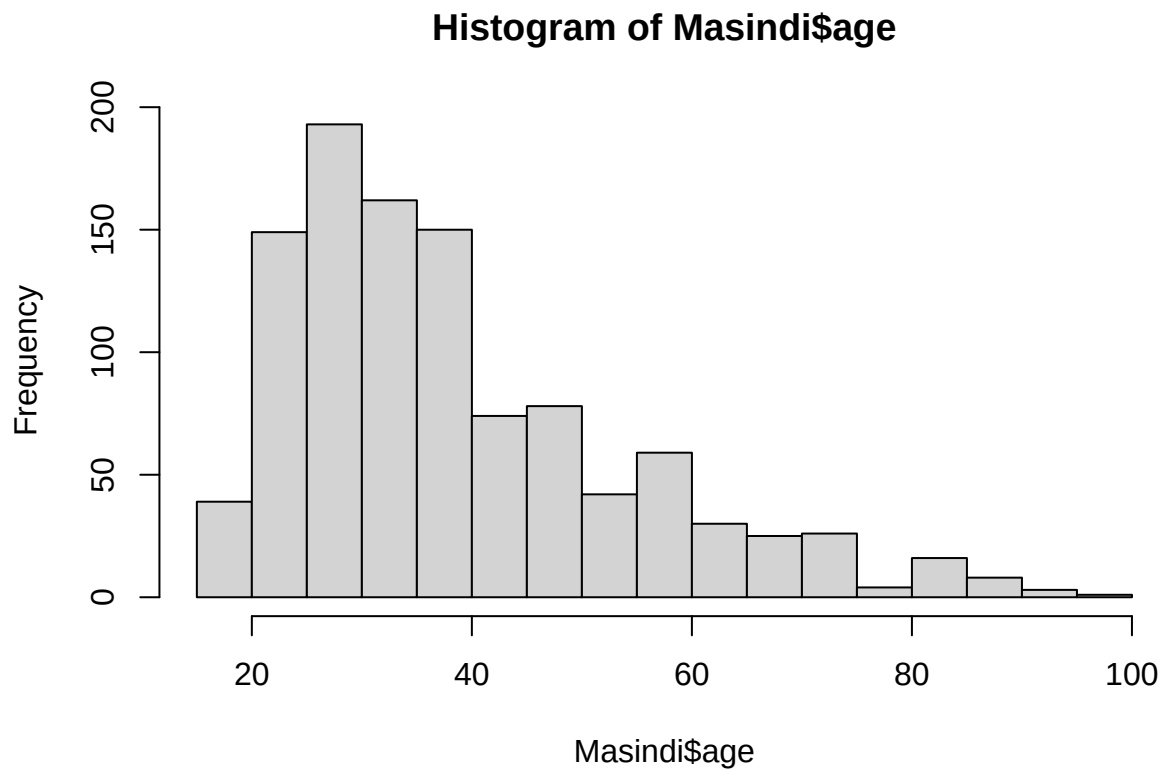
```



```
# dev.off()
```

Next, the relationship between age and sex was statistically assessed.

```
# AGE and SEX Visualizing Age and Sex  
hist(Masindi$age)
```



```
Masindi_Male <- Masindi[Masindi$sex == "M", ]
Masindi_Female <- Masindi[Masindi$sex == "F", ]

# tiff('FigB.tiff', units='px', width=2200, height=2000,
# res=400)
par(mfrow = c(1, 2))
hist(Masindi_Male$age, xlab = "Age (years)", main = "Male", col = "gold2",
     ylim = c(0, 250))
# NOT normally distributed
hist(Masindi_Female$age, xlab = "Age (years)", main = "Female",
     col = "red")
```



```
# NOT normally distributed
par(mfrow = c(1, 1))
# dev.off()

# Used non-parametric Mann-Whitney-Wilcoxon test rather
# than t-test
wilcox.test(Masindi_Male$age, Masindi_Female$age, conf.int = T)
```

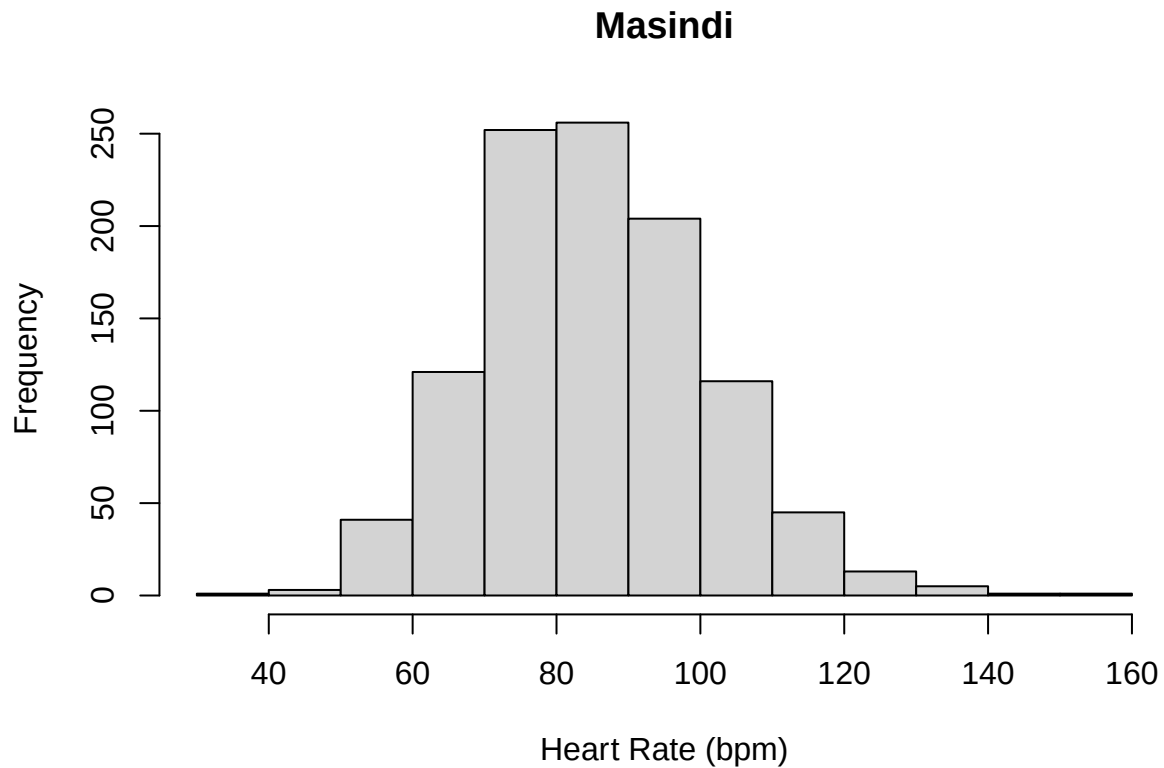
```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_Male$age and Masindi_Female$age
## W = 169406, p-value = 3.176e-13
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
## 4.000016 7.000043
## sample estimates:
## difference in location
## 5.999962
```

```
# p < .01, Reject the null, There is a significant
# difference between the age of males and females
```

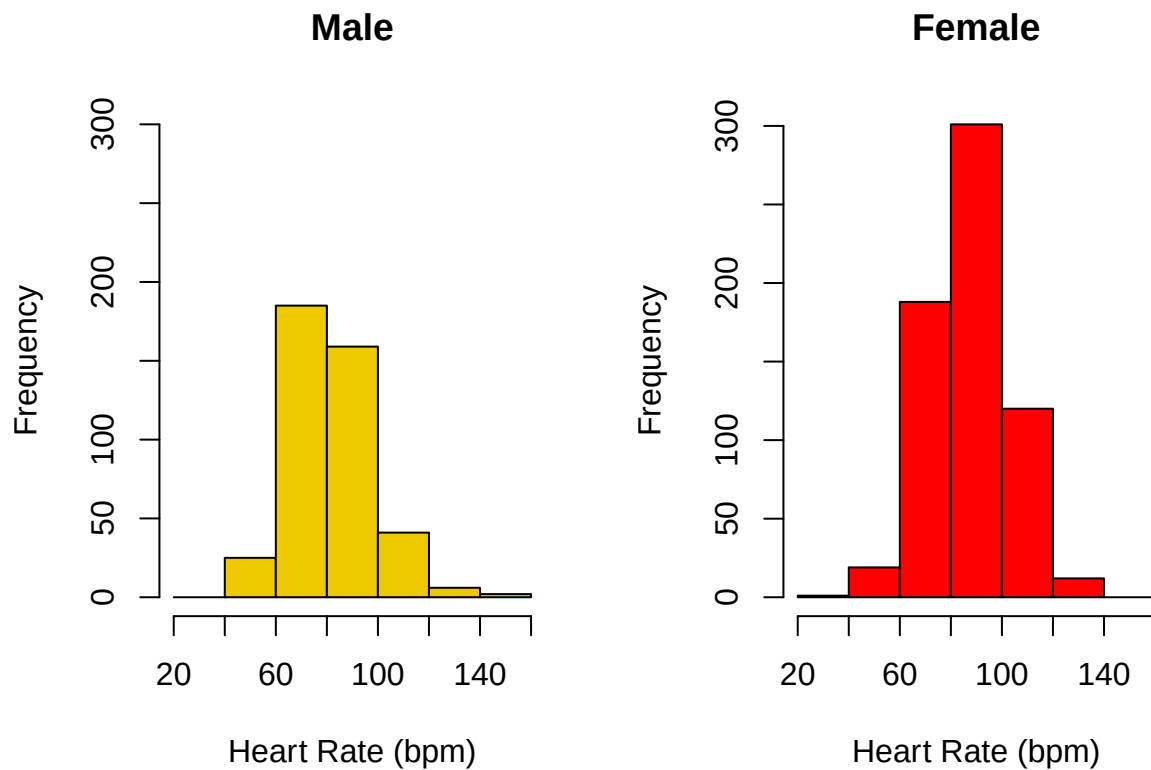
According to the non-parametric Mann-Whitney Wilcoxon test, there is a significant difference ($p < .01$) between the ages of males and females in the Masindi cohort.

The relationship between heart rate and sex was then assessed.

```
# HEART RATE and SEX Visualizing Heart Rate and Sex
hist(Masindi$hr, xlab = "Heart Rate (bpm)", main = "Masindi") # Normally Distributed
```



```
# tiff('FigC.tiff', units='px', width=2200, height=2000,
# res=400)
par(mfrow = c(1, 2))
hist(Masindi_Male$hr, xlab = "Heart Rate (bpm)", main = "Male",
     col = "gold2", breaks = seq(from = 20, to = 160, by = 20),
     ylim = c(0, 300)) # Possibly normally distributed?
hist(Masindi_Female$hr, xlab = "Heart Rate (bpm)", main = "Female",
     col = "red", breaks = seq(from = 20, to = 160, by = 20)) # Normally distributed
```

```
par(mfrow = c(1, 1))
# dev.off()

# Confirmation with Shapiro-Wilk test
with(Masindi_Male, shapiro.test(hr)) # p < .01, Reject the Null, NOT normally distributed

##
## Shapiro-Wilk normality test
##
## data: hr
## W = 0.96449, p-value = 1.601e-08

with(Masindi_Female, shapiro.test(hr)) # p > .05, Fail to Reject the Null, Normally distributed

##
## Shapiro-Wilk normality test
##
## data: hr
## W = 0.99649, p-value = 0.1718

wilcox.test(Masindi_Male$hr, Masindi_Female$hr, conf.int = T)

##
```

```
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_Male$hr and Masindi_Female$hr
## W = 100847, p-value = 9.776e-12
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
## -8.000046 -4.999980
## sample estimates:
## difference in location
## -6.999951
```

```
# p < .01, Reject the null, There is a significant
# difference between the hr of males and females

print(c(mean(Masindi_Female$hr), sd(Masindi_Female$hr)))
```

```
## [1] 88.20125 14.80272
```

```
print(c(mean(Masindi_Male$hr), sd(Masindi_Male$hr)))
```

```
## [1] 82.22249 15.89416
```

```
mean(Masindi_Female$hr) - mean(Masindi_Male$hr)
```

```
## [1] 5.97876
```

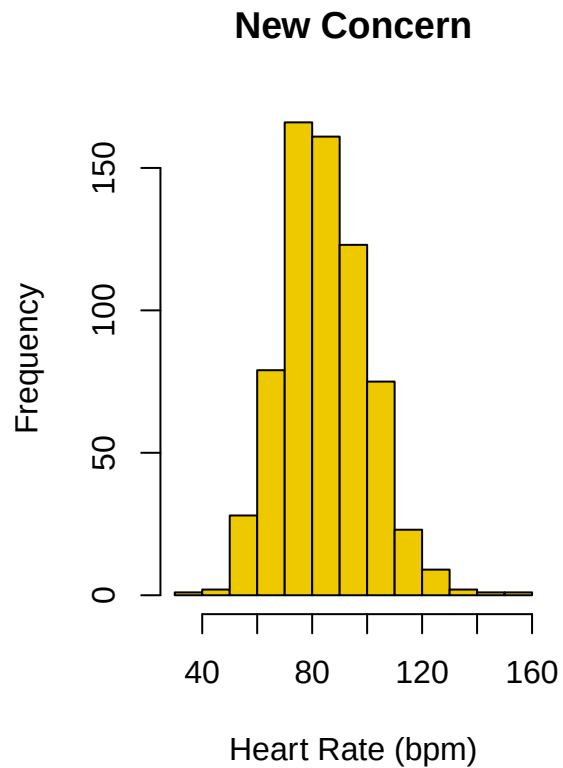
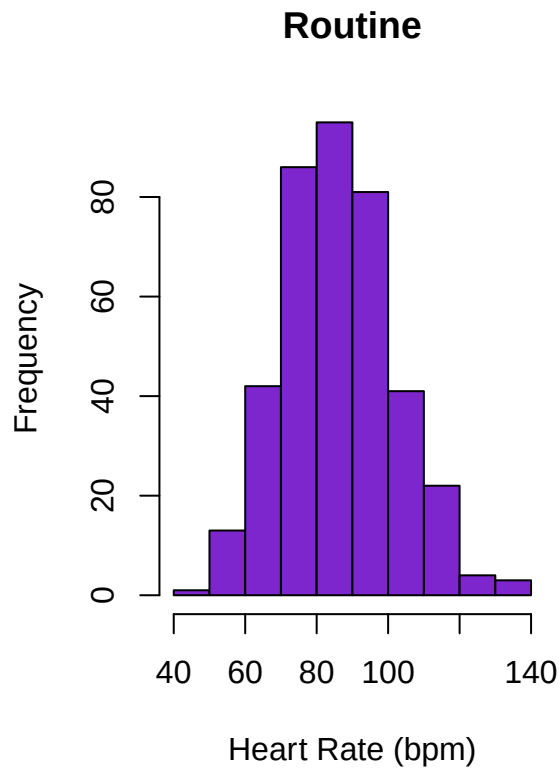
According to the Mann-Whitney Wilcoxon test, there is a significant difference ($p < .01$) between the heart rate of males and females. Without age adjustment, the mean heart rate of females (88.20 ± 14.80) was 5.98 bpm higher than in males (82.22 ± 15.89).

The relationship between heart rate and reason for visit was then assessed.

```
# HEART RATE and REASON for visit

Masindi_newconcern <- Masindi[Masindi$reason == "New Concern",
]

par(mfrow = c(1, 2))
hist(Masindi_routine$hr, xlab = "Heart Rate (bpm)", main = "Routine",
     col = "purple3") # Normally distributed
hist(Masindi_newconcern$hr, xlab = "Heart Rate (bpm)", main = "New Concern",
     col = "gold2") # Normally distributed
```



```
par(mfrow = c(1, 1))

# Welch's two sample, two-tailed t-test
t.test(Masindi_routine$hr, Masindi_newconcern$hr, conf.int = T)

##
##  Welch Two Sample t-test
##
## data:  Masindi_routine$hr and Masindi_newconcern$hr
## t = 1.605, df = 817.81, p-value = 0.1089
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.3524505  3.5134274
## sample estimates:
## mean of x mean of y
##  86.84278  85.26230

# NOT significant
```

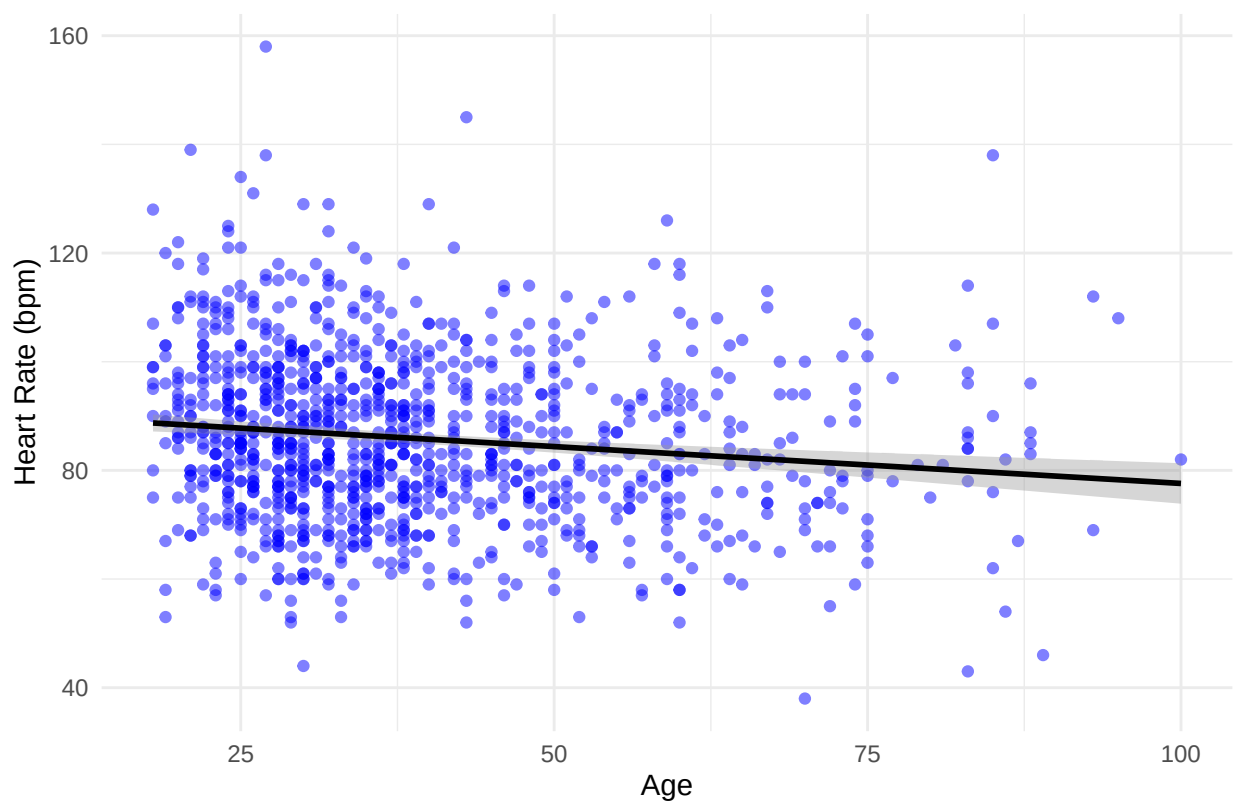
The heart rates for both groups are normally distributed, so a t-test was performed. The test showed no significant difference between the heart rates of patients presenting with new concerns and those presenting for routine visits.

Next, the relationship between heart rate and age was assessed.

```
#HEART RATE and AGE
```

```
plot1 <- ggplot(Masindi, aes(x = age, y = hr)) +  
  geom_point(color = "blue", alpha = 0.5) + # Scatterplot points  
  geom_smooth(method = "lm", color = "black", se = TRUE) + # Regression line  
  labs(title = "", x = "Age", y = "Heart Rate (bpm)") +  
  theme_minimal()  
  
plot1
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
# Data points appear linear with hr decreasing with age
```

```
# Linear regression model
```

```
hr_age_model <- lm(hr ~ age, data = Masindi)
```

```
summary(hr_age_model) # p < .01, reject null
```

```
##
```

```
## Call:
```

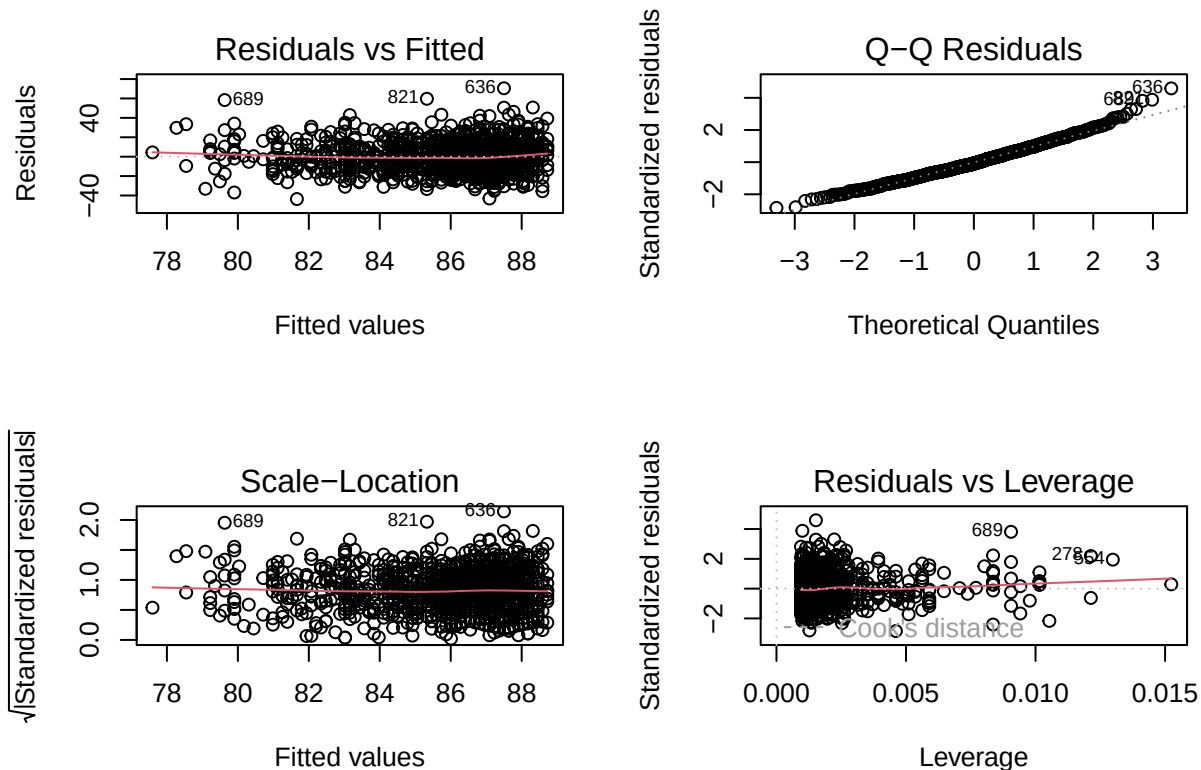
```
## lm(formula = hr ~ age, data = Masindi)
```

```
##
```

```
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -43.664 -10.372  -1.279   9.986  70.500
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  91.16445    1.27693   71.394 < 2e-16 ***
## age         -0.13571    0.03025   -4.487 8.01e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.38 on 1057 degrees of freedom
## Multiple R-squared:  0.01869,    Adjusted R-squared:  0.01776
## F-statistic: 20.13 on 1 and 1057 DF,  p-value: 8.01e-06
```

```
# Check Assumptions
par(mfrow = c(2, 2))
plot(hr_age_model)
```



```
par(mfrow = c(1, 1))
# Q-Q plot shows normality of residuals
# Scale-Location shows homoskedasticity
# Few extreme outliers in Residuals vs. Leverage
```

The linear regression model showed a significant relationship ($p < .01$) between heart rate and age. Without adjustment for sex, heart rate was found to decrease by 0.136 bpm per year of age. The model assumptions were met, indicating that the model was appropriate for the data.

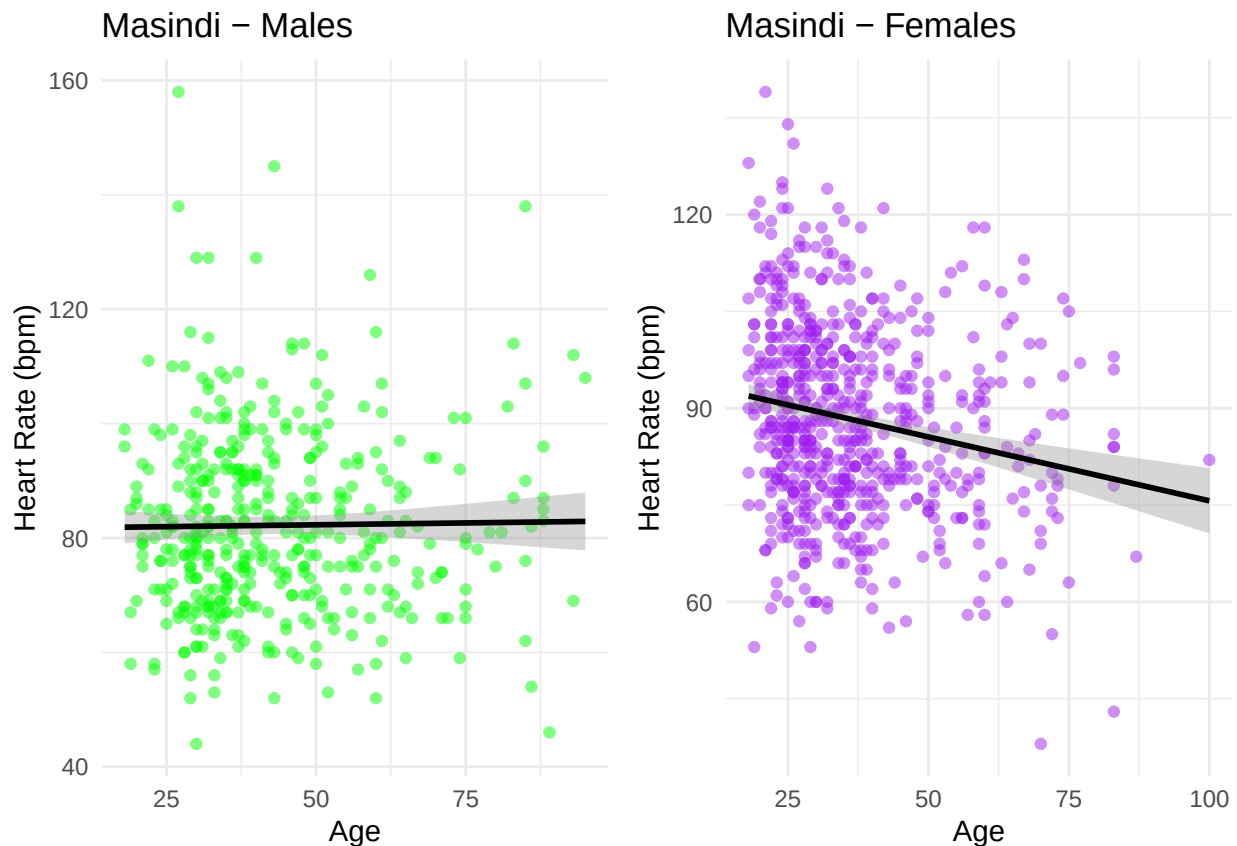
Finally, the relationship between heart rate and both age and sex was evaluated.

```
# HEART RATE, AGE, and SEX Visualizing
p2 <- ggplot(Masindi_Male, aes(x = age, y = hr)) + geom_point(color = "green",
  alpha = 0.5) + geom_smooth(method = "lm", color = "black",
  se = TRUE) + labs(title = "Masindi - Males", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal()

p3 <- ggplot(Masindi_Female, aes(x = age, y = hr)) + geom_point(color = "purple",
  alpha = 0.5) + geom_smooth(method = "lm", color = "black",
  se = TRUE) + labs(title = "Masindi - Females", x = "Age",
  y = "Heart Rate (bpm)") + theme_minimal()

gridExtra::grid.arrange(p2, p3, nrow = 1, ncol = 2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```



```
# Apparent interaction between sex and age on heart rate
# Age and HR appear unrelated in males, while HR appears to
# decrease with age in females
```

The scatterplots show an interaction between age and sex on heart rate. This interaction was then tested below. Because the outcome variable (`hr`) is continuous and the predictor variables are both continuous (`age`) and categorical (`sex`), an analysis of covariance (ANCOVA) model was used to test for interaction.

```
# ANCOVA model (w/ interaction)
additive_model <- aov(hr ~ age + sex, data = Masindi)

interaction_model <- aov(hr ~ age * sex, data = Masindi)
summary(interaction_model) # p < .01 for interaction term
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## age           1   4760     4760   20.90 5.41e-06 ***
## sex           1   6872     6872   30.18 4.95e-08 ***
## age:sex       1   2740     2740   12.03 0.000544 ***
## Residuals    1055 240254         228
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(additive_model, interaction_model) # p < .01, interaction term significantly improves model
```

```
## Analysis of Variance Table
##
## Model 1: hr ~ age + sex
## Model 2: hr ~ age * sex
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     1056 242994
## 2     1055 240254   1    2740.3 12.033 0.0005437 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The interaction term was found to be significant ($p < .01$) and the model was significantly improved by its inclusion rather than only additive effects ($p < .01$).

Separate linear regression models were then used to assess the relationship between age and heart rate in males and females.

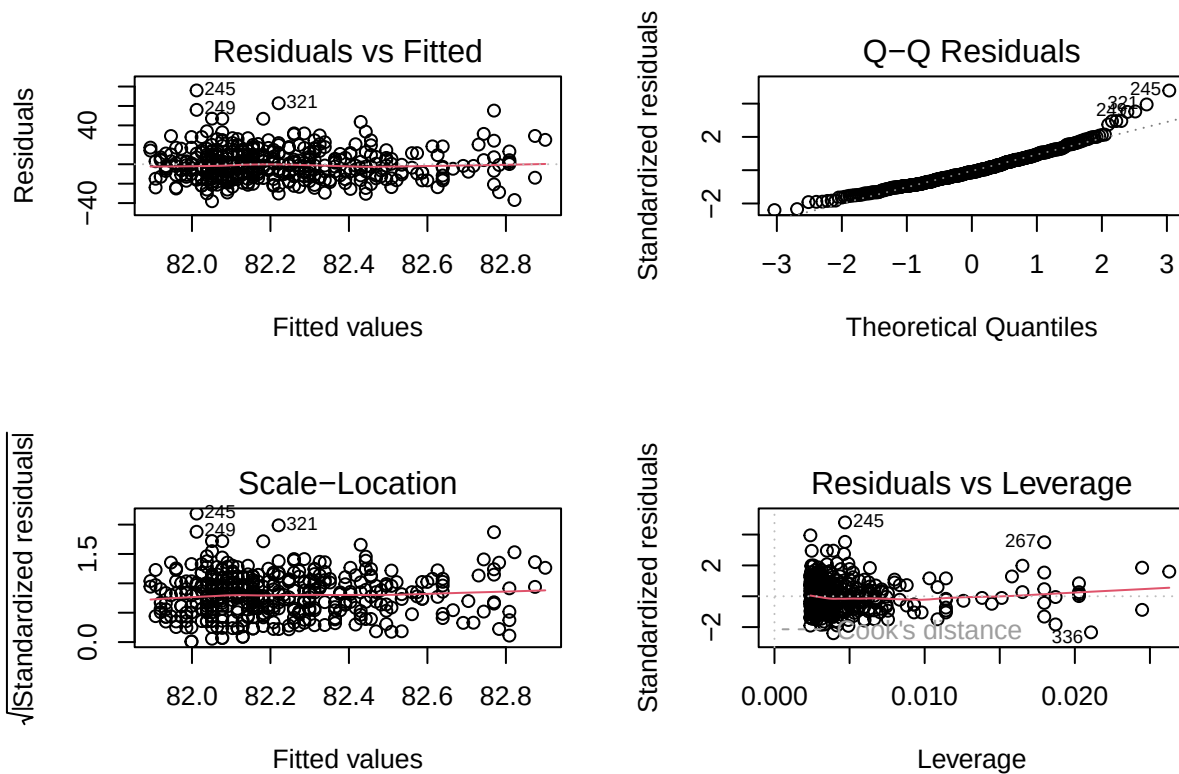
```
# Separate linear regression models (hr ~ age) for males
# and females

# MALE
male_model <- lm(hr ~ age, data = Masindi_Male)
summary(male_model) # p > .05, No significant affect of age on hr in males
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_Male)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -38.051 -11.146  -1.947   9.854  75.988
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 81.65907    2.18760  37.328  <2e-16 ***
## age         0.01307    0.04741   0.276    0.783
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.91 on 416 degrees of freedom
## Multiple R-squared:  0.0001825, Adjusted R-squared:  -0.002221
## F-statistic: 0.07595 on 1 and 416 DF,  p-value: 0.783
```

```
# Test linear regression assumptions
par(mfrow = c(2, 2))
plot(male_model)
```



```
par(mfrow = c(1, 1))
# Assumptions met: Linearity, normality of residuals,
# homoscedasticity

# FEMALE
female_model <- lm(hr ~ age, data = Masindi_Female)
summary(female_model) # p < .01, Significant affect of age on hr in females
```

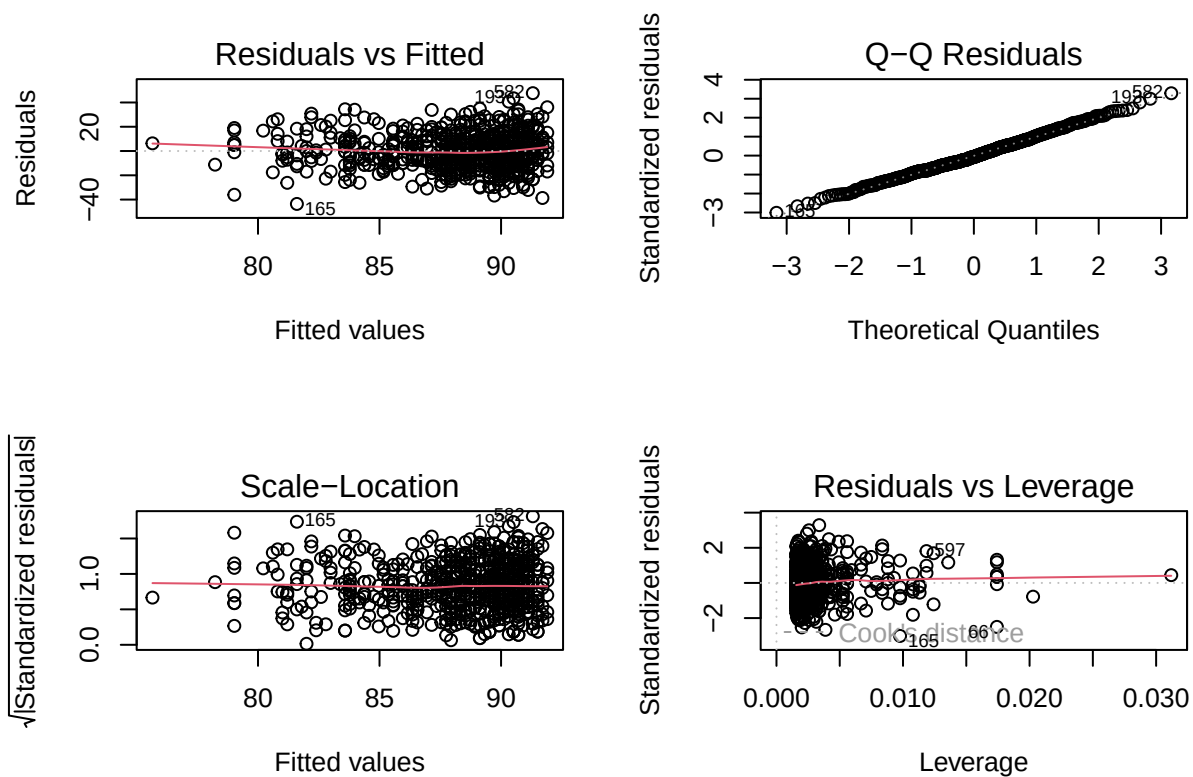
```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_Female)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -43.601  -9.959  -1.026   9.664  47.693
```



```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 95.46718    1.55867   61.249 < 2e-16 ***
## age        -0.19809    0.03951  -5.014 6.92e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.53 on 639 degrees of freedom
## Multiple R-squared:  0.03785,    Adjusted R-squared:  0.03635
## F-statistic: 25.14 on 1 and 639 DF,  p-value: 6.917e-07
```

```
# In females, HR decreases by 0.20 bpm for each year
# increase in age

# Test linear regression assumptions
par(mfrow = c(2, 2))
plot(female_model)
```



```
par(mfrow = c(1, 1))
# Assumptions met: Linearity, normality of residuals,
# homoscedasticity
```

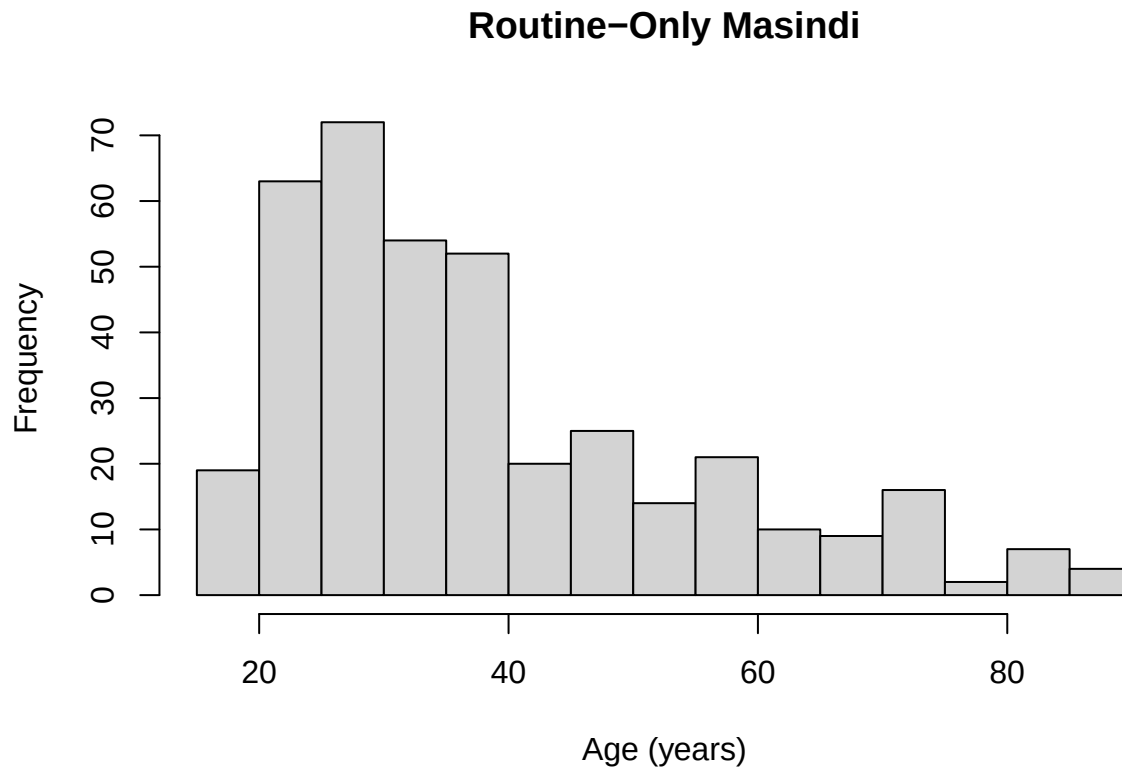
The linear regression models indicated that age does not significantly affect heart rates in males ($p > .05$), while it significantly decreases heart rates in females ($p < .01$) by 0.20 bpm for each year in age. The assumptions of linear regression were once again tested.

Routine Visits Only

The code below repeats the analysis, excluding patients presenting with new concerns (“routine only”).

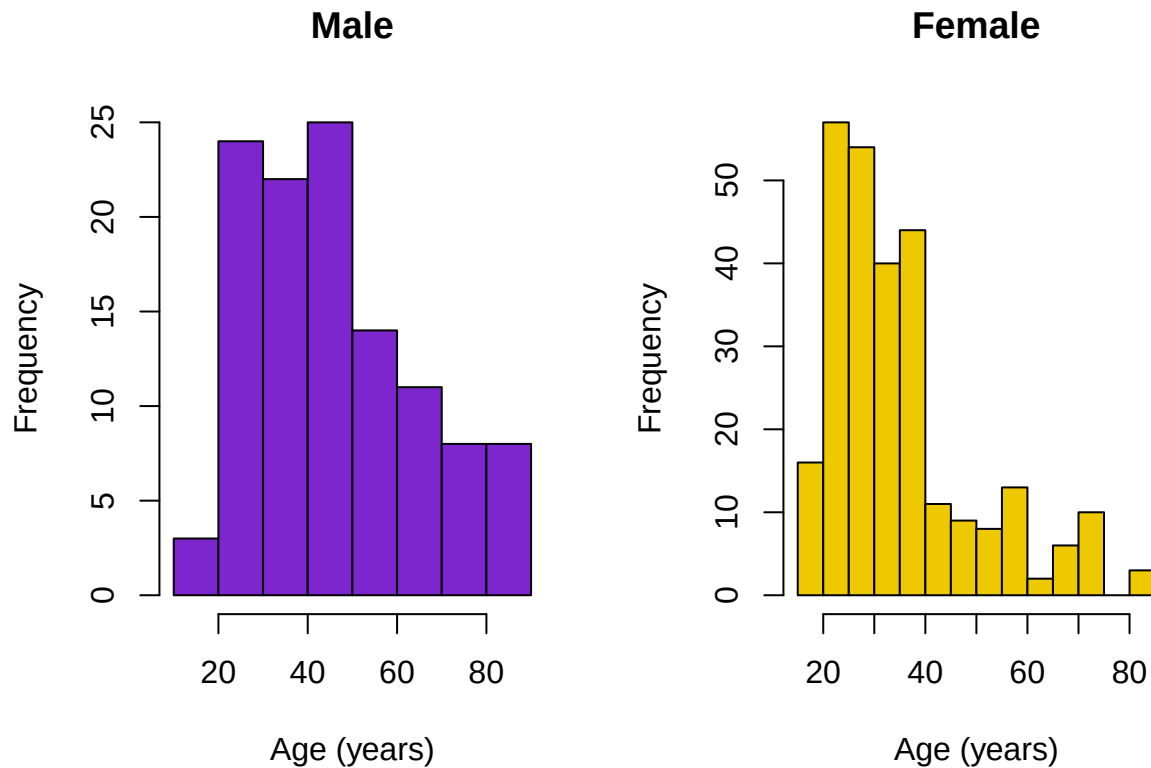
```
#### Repeat analysis EXCLUDING patients presenting with new
#### concerns####

# AGE and SEX Visualizing Age and Sex
hist(Masindi_routine$Age, xlab = "Age (years)", main = "Routine-Only Masindi")
```



```
Masindi_routine_Male <- Masindi_routine[Masindi_routine$sex ==
  "M", ]
Masindi_routine_Female <- Masindi_routine[Masindi_routine$sex ==
  "F", ]

par(mfrow = c(1, 2))
hist(Masindi_routine_Male$Age, xlab = "Age (years)", main = "Male",
  col = "purple3") # NOT normally distributed
hist(Masindi_routine_Female$Age, xlab = "Age (years)", main = "Female",
  col = "gold2") # NOT normally distributed
```



```
par(mfrow = c(1, 1))

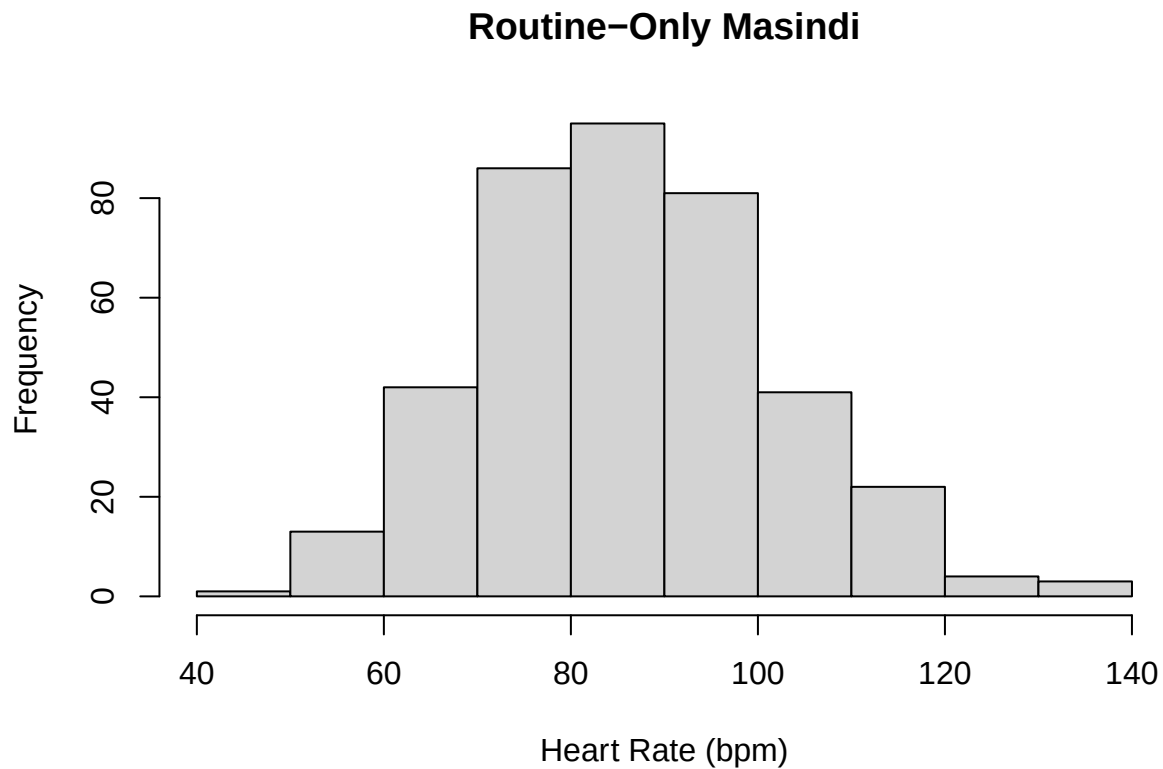
# Used non-parametric Mann-Whitney-Wilcoxon test rather
# than t-test
wilcox.test(Masindi_routine_Male$Age, Masindi_routine_Female$Age,
  conf.int = T)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_routine_Male$Age and Masindi_routine_Female$Age
## W = 22107, p-value = 2.074e-10
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
## 7.00006 14.00008
## sample estimates:
## difference in location
## 10
```

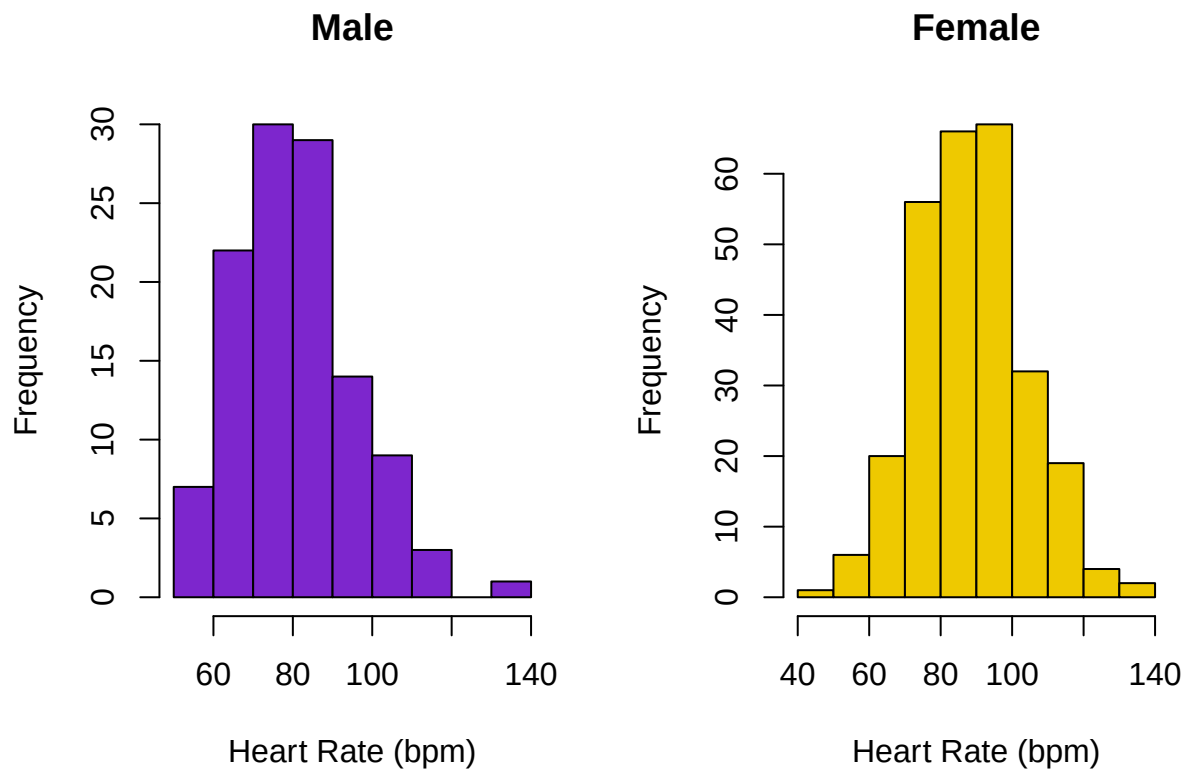
```
# p < .01, Reject the null, There is a significant
# difference between the age of males and females
```

Significant difference ($p < .01$) between the age of males and females remains after excluding “new concern” patients.

```
# HEART RATE and SEX Visualizing Heart Rate and Sex
hist(Masindi_routine$hr, xlab = "Heart Rate (bpm)", main = "Routine-Only Masindi") # Normally Distributed
```



```
par(mfrow = c(1, 2))
hist(Masindi_routine_Male$hr, xlab = "Heart Rate (bpm)", main = "Male",
     col = "purple3") # Not normally distributed
hist(Masindi_routine_Female$hr, xlab = "Heart Rate (bpm)", main = "Female",
     col = "gold2") # Possibly normally distributed?
```



```
par(mfrow = c(1, 1))

# Use non-parametric alternative to t-test
wilcox.test(Masindi_routine_Male$hr, Masindi_routine_Female$hr,
  conf.int = T)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_routine_Male$hr and Masindi_routine_Female$hr
## W = 11040, p-value = 3.874e-06
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
## -11.000045 -4.999959
## sample estimates:
## difference in location
## -7.999943
```

```
# p < .01, Reject the null, There is a significant
# difference between the hr of males and females

print(c(mean(Masindi_routine_Female$hr), sd(Masindi_routine_Female$hr)))
```

```
## [1] 89.06960 15.08964
```

```
print(c(mean(Masindi_routine_Male$hr), sd(Masindi_routine_Male$hr)))
```

```
## [1] 81.55652 14.73401
```

```
mean(Masindi_routine_Female$hr) - mean(Masindi_routine_Male$hr)
```

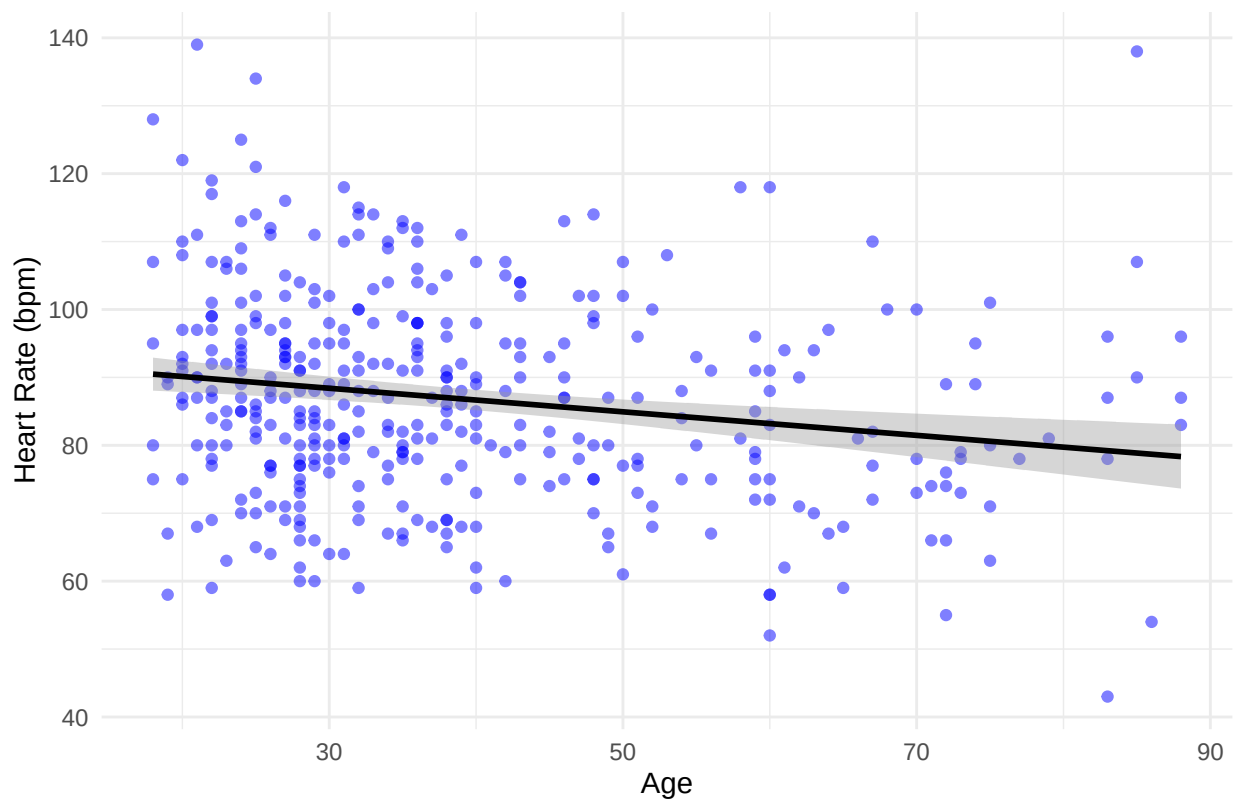
```
## [1] 7.513075
```

Significant difference ($p < .01$, 7.51 bpm) between the heart rate of females (89.07 ± 15.09) and males (81.56 ± 14.73).

```
#HEART RATE and AGE
```

```
plot1 <- ggplot(Masindi_routine, aes(x = age, y = hr)) +  
  geom_point(color = "blue", alpha = 0.5) + # Scatterplot points  
  geom_smooth(method = "lm", color = "black", se = TRUE) + # Regression line  
  labs(title = "", x = "Age", y = "Heart Rate (bpm)") +  
  theme_minimal()  
  
plot1
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```

# Data points appear linear with hr decreasing with age

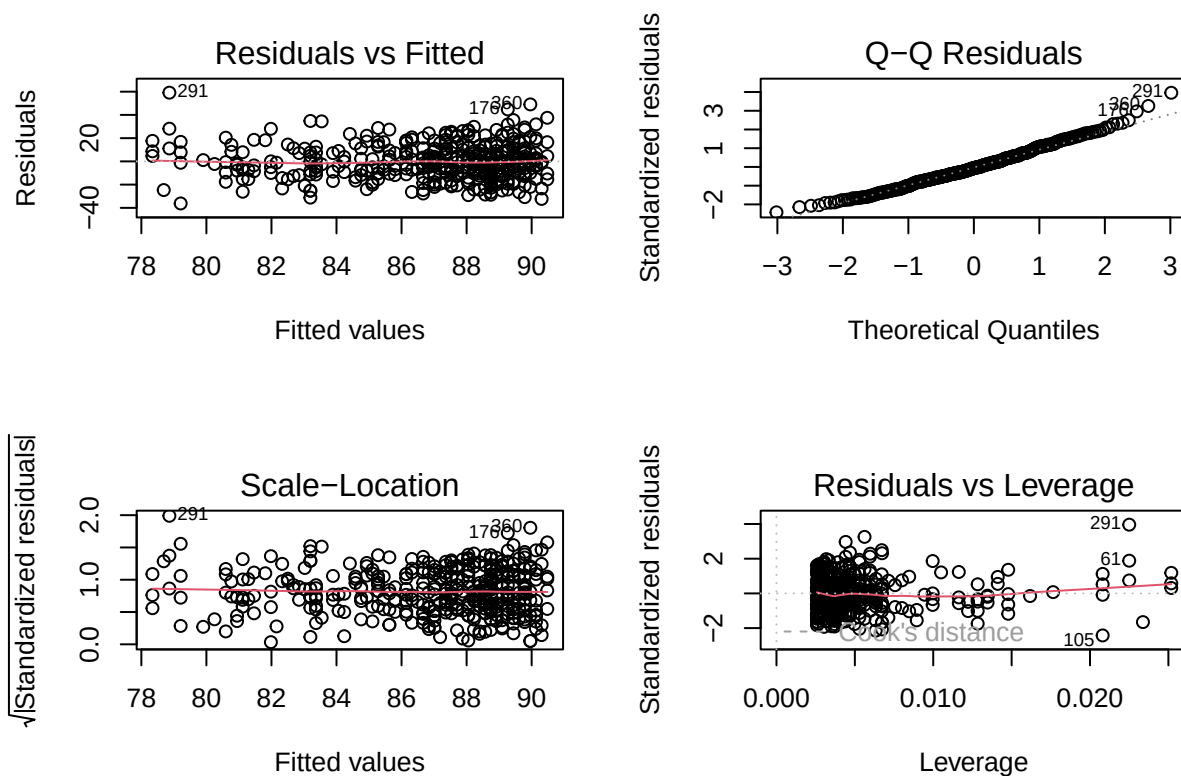
# Linear regression model
hr_age_routine_model <- lm(hr ~ age, data = Masindi_routine)

summary(hr_age_routine_model) # p < .01, reject null

##
## Call:
## lm(formula = hr ~ age, data = Masindi_routine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -36.209 -10.044  -0.925   9.213  59.138
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  93.60190    1.96157  47.718  < 2e-16 ***
## age         -0.17341    0.04632  -3.744  0.000209 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.1 on 386 degrees of freedom
## Multiple R-squared:  0.03504,    Adjusted R-squared:  0.03254
## F-statistic: 14.01 on 1 and 386 DF,  p-value: 0.000209

# Check Assumptions
par(mfrow = c(2, 2))
plot(hr_age_routine_model)

```



```
par(mfrow = c(1, 1))
# Q-Q plot shows normality of residuals
# Scale-Location shows homoskedasticity
# Few extreme outliers in Residuals vs. Leverage
```

Significant relationship ($p < .01$) remains between heart rate and age in the Masindi routine cohort. Heart rate decreases by 0.17 bpm per year of age.

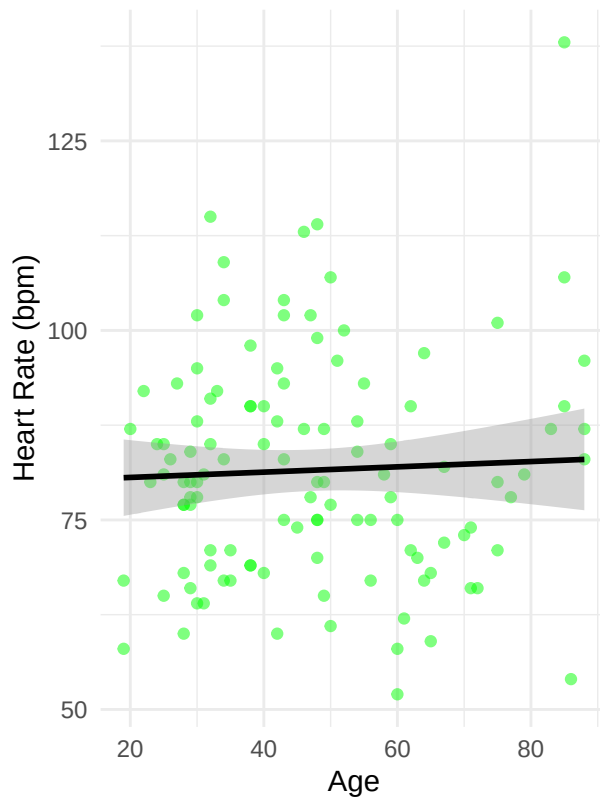
```
# HEART RATE, AGE, and SEX Visualizing
p2 <- ggplot(Masindi_routine_Male, aes(x = age, y = hr)) + geom_point(color = "green",
  alpha = 0.5) + geom_smooth(method = "lm", color = "black",
  se = TRUE) + labs(title = "Masindi (Routine) - Males", x = "Age",
  y = "Heart Rate (bpm)") + theme_minimal()

p3 <- ggplot(Masindi_routine_Female, aes(x = age, y = hr)) +
  geom_point(color = "purple", alpha = 0.5) + geom_smooth(method = "lm",
  color = "black", se = TRUE) + labs(title = "Masindi (Routine) - Females",
  x = "Age", y = "Heart Rate (bpm)") + theme_minimal()

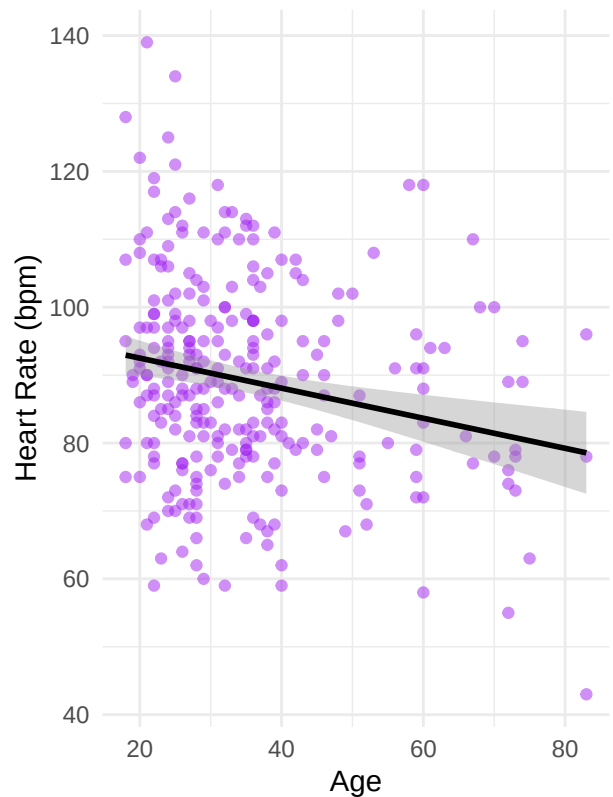
gridExtra::grid.arrange(p2, p3, nrow = 1, ncol = 2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```


Masindi (Routine) – Males



Masindi (Routine) – Females



```
# Apparent interaction between sex and age on heart rate HR
# appears to slightly increase w/ age in males, decrease
# with age in females

# ANCOVA model (w/ interaction)
additive_routine_model <- aov(hr ~ age + sex, data = Masindi_routine)

interaction_routine_model <- aov(hr ~ age * sex, data = Masindi_routine)
summary(interaction_routine_model) # p < .01 for interaction term
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## age         1   3197     3197   14.643 0.000152 ***
## sex         1   2717     2717   12.443 0.000470 ***
## age:sex      1    1500     1500    6.872 0.009104 **
## Residuals   384  83835         218
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(additive_routine_model, interaction_routine_model) # p < .01, interaction term significantly imp
```

```
## Analysis of Variance Table
##
## Model 1: hr ~ age + sex
## Model 2: hr ~ age * sex
##   Res.Df  RSS Df Sum of Sq    F Pr(>F)
```

```
## 1      385 85336
## 2      384 83835  1      1500.3 6.872 0.009104 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

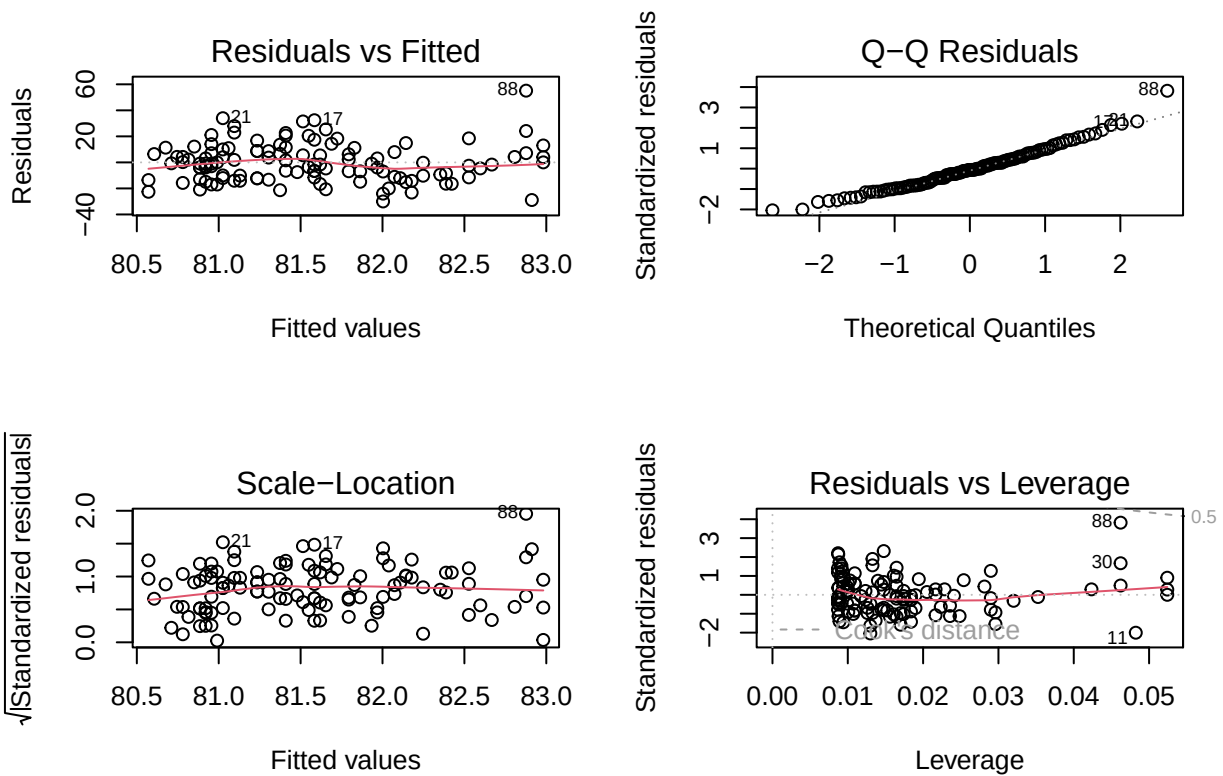
Once again, ANCOVA demonstrates that the interaction of age and sex significantly improves the model for heart rate ($p < .01$).

```
# Separate linear regression models (hr ~ age) for males
# and females

# MALE
male_routine_model <- lm(hr ~ age, data = Masindi_routine_Male)
summary(male_routine_model) # p > .05, No significant affect of age on hr in males
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_routine_Male)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.003 -11.556  -0.934   8.730  55.123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  79.90731    3.83420  20.841  <2e-16 ***
## age          0.03493    0.07579   0.461    0.646
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.79 on 113 degrees of freedom
## Multiple R-squared:  0.001877,    Adjusted R-squared:  -0.006956
## F-statistic: 0.2125 on 1 and 113 DF,  p-value: 0.6457
```

```
# Test linear regression assumptions
par(mfrow = c(2, 2))
plot(male_routine_model)
```



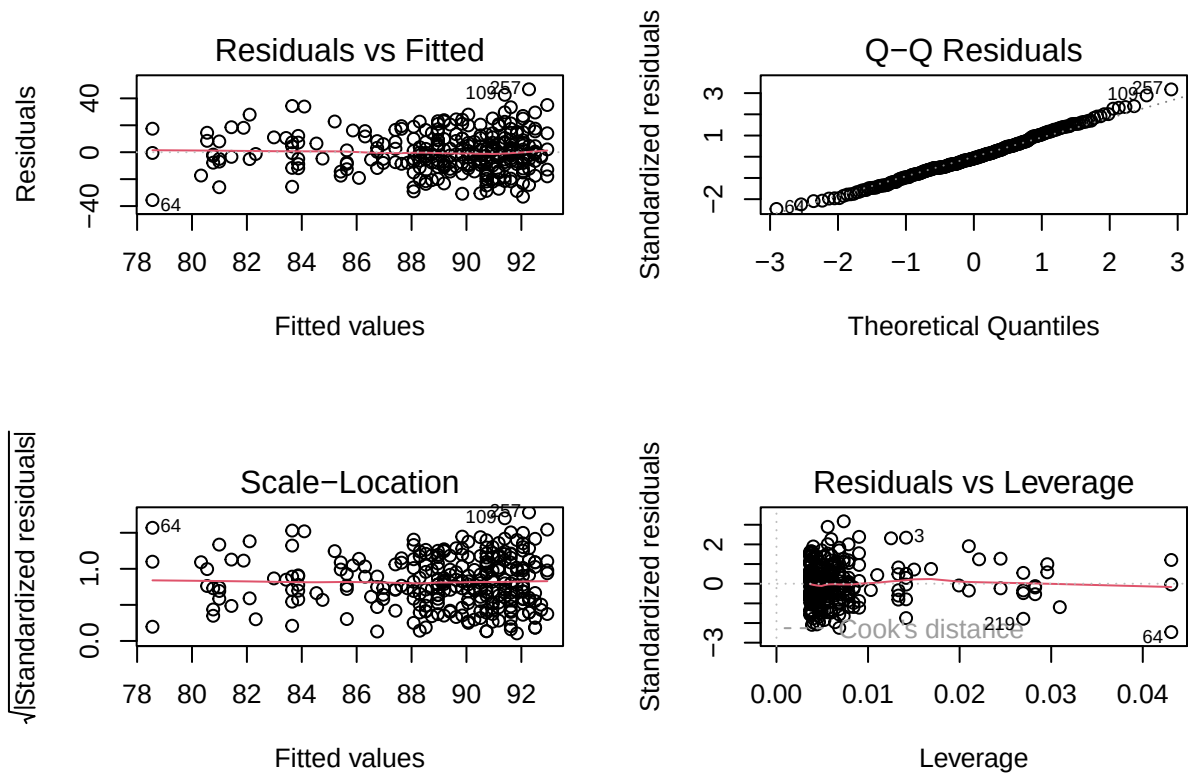
```
par(mfrow = c(1, 1))
# Assumptions met: Linearity, normality of residuals,
# homoscedasticity

# FEMALE
female_routine_model <- lm(hr ~ age, data = Masindi_routine_Female)
summary(female_routine_model) # p < .01, Significant affect of age on hr in females
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_routine_Female)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.559  -9.182  -1.174   9.039  46.719
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  96.92806    2.36890  40.917  < 2e-16 ***
## age         -0.22131    0.06178  -3.582  0.000404 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.77 on 271 degrees of freedom
## Multiple R-squared:  0.04521,    Adjusted R-squared:  0.04169
## F-statistic: 12.83 on 1 and 271 DF,  p-value: 0.0004037
```

```
# In females, HR decreases by 0.22 bpm for each year
# increase in age

# Test linear regression assumptions
par(mfrow = c(2, 2))
plot(female_routine_model)
```



```
par(mfrow = c(1, 1))
# Assumptions met: Linearity, normality of residuals,
# homoscedasticity
```

As before, age does not significantly affect heart rate in males ($p > .05$), while it significantly decreases heart rate in females ($p < .01$) by 0.22 bpm per year. The assumptions of linear regression were met.

Masindi_Census_Comparison.R

To assess the generalizability of findings to the local region, the previous ANCOVA results, along with census projection data, were used to calculate the district-level mean expected heart rate for Masindi.

First, the 2022 census data, including the estimated male and female population counts for each year of age, was reformatted to match the Masindi cohort dataset (`Masindi_data.csv`).

```
library(tidyverse)
```

```
folder_path <- as.character(here::here()) #>>>THIS SHOULD POINT TO THE "MAIN" FOLDER<<<<#  
Masindi <- read_csv(paste0(folder_path, "/Processed Data/Masindi/Masindi_data.csv"))
```

```
## Rows: 1059 Columns: 6  
## -- Column specification -----  
## Delimiter: ","  
## chr (3): sex, reason, source  
## dbl (3): Masindi_ID, hr, age  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

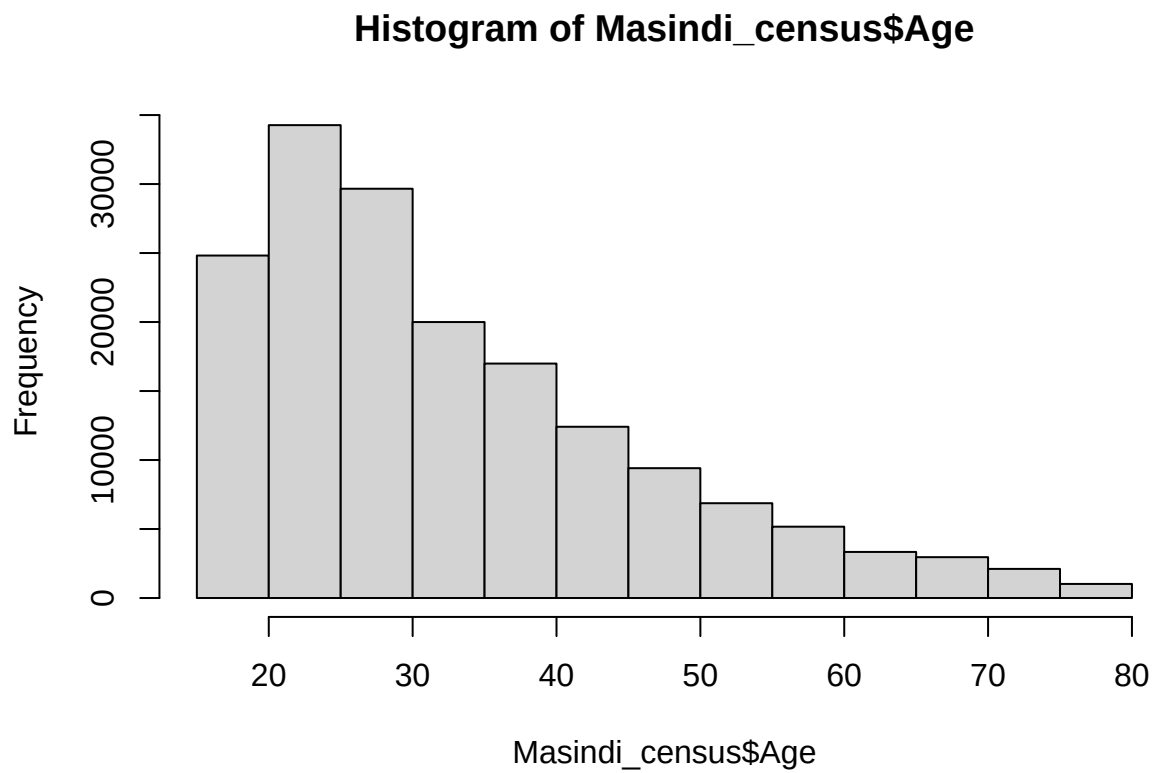
```
Masindi_census <- read_csv(paste0(folder_path, "/Processed Data/Masindi/Masindi_Census_Projections.csv"))
```

```
## Rows: 80 Columns: 3  
## -- Column specification -----  
## Delimiter: ","  
## dbl (1): Age  
## num (2): Male, Female  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

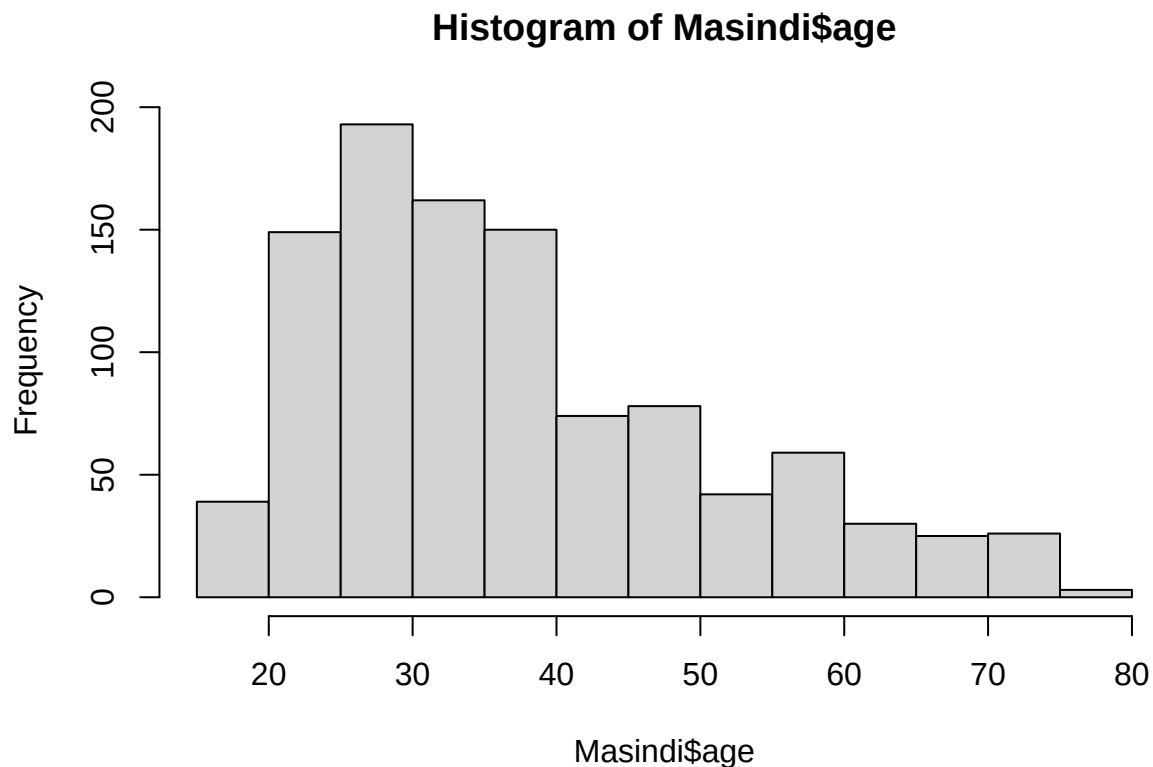
```
Masindi <- Masindi %>% #To Match Masindi census data  
  filter(age < 80)  
  
Masindi_census <- pivot_longer( #To Match formatting of cohort data  
  Masindi_census,  
  cols = c(Male, Female),  
  names_to = "sex",  
  values_to = "Count"  
) %>%  
  drop_na (Count) %>%  
  uncount(weights = Count) %>%  
  mutate(sex = case_when(  
    sex == "Male" ~ "M",  
    sex == "Female" ~ "F",  
    TRUE ~ sex))
```

The age distribution of the clinical cohort was compared to that of the entire district population.

```
hist(Masindi_census$Age) # NOT normally distributed
```



```
hist(Masindi$Age) # NOT normally distributed
```



```
# Used non-parametric Mann-Whitney-Wilcoxon test rather
# than t-test for age
wilcox.test(Masindi$age, Masindi_census$Age) # p < .001
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi$age and Masindi_census$Age
## W = 105433145, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

Next, the sex ratio of the clinical cohort was compared to that of the entire district population.

```
# Sex comparison
table_sex <- table( # Contingency table
  factor(c(Masindi_census$sex, Masindi$sex), levels = c("M", "F")),
  factor(c(rep("Masindi Census", length(Masindi_census$sex)), rep("Masindi", length(Masindi$sex))), levels = c("M", "F"))
)

chi_square_test <- chisq.test(table_sex)

print(chi_square_test)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
```

```
##
## data: table_sex
## X-squared = 65.947, df = 1, p-value = 4.633e-16
```

```
# p < .001
```

Then, the ages distributions were again compared; this time separately for males and females.

```
# Age, separated by sex
MCL_Male <- Masindi_census[Masindi_census$sex == "M", ]
MCL_Female <- Masindi_census[Masindi_census$sex == "F", ]
Masindi_Male <- Masindi[Masindi$sex == "M", ]
Masindi_Female <- Masindi[Masindi$sex == "F", ]

wilcox.test(Masindi_Male$Age, MCL_Male$Age) # p < .001
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_Male$Age and MCL_Male$Age
## W = 23115555, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(Masindi_Female$Age, MCL_Female$Age) # p < .001
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Masindi_Female$Age and MCL_Female$Age
## W = 29511905, p-value = 3.313e-09
## alternative hypothesis: true location shift is not equal to 0
```

```
proportion_male <- nrow(MCL_Male)/nrow(Masindi_census)
proportion_female <- nrow(MCL_Female)/nrow(Masindi_census)

mean_age_male <- mean(MCL_Male$Age)
mean_age_female <- mean(MCL_Female$Age)
```

For both sexes, ages distributions differ significantly between the cohort and district population.

Using the coefficients from the ANCOVA model selected in `Masindi_HR_Analysis.R` (complete dataset), the expected heart rate was calculated for the district-level demographic structure.

The model was reproduced and coefficients extracted.

```
model <- aov(hr ~ age * sex, data = Masindi) # From 'Masindi_HR_Analysis.R'
summary.lm(model)
```

```
##
## Call:
```



```
## aov(formula = hr ~ age * sex, data = Masindi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -43.411 -10.458  -1.287   9.665  75.370
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  95.63285    1.69991  56.258 < 2e-16 ***
## age         -0.20317    0.04427  -4.590 4.99e-06 ***
## sexM        -11.67579    2.92468  -3.992 7.01e-05 ***
## age:sexM      0.15403    0.07074   2.177  0.0297 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.91 on 1026 degrees of freedom
## Multiple R-squared:  0.0609, Adjusted R-squared:  0.05815
## F-statistic: 22.18 on 3 and 1026 DF,  p-value: 6.436e-14
```

```
# Extract coefficients
beta_0 <- coef(model)[1] # Intercept
beta_1 <- coef(model)[2] # Age effect
beta_2 <- coef(model)[3] # Sex effect
beta_3 <- coef(model)[4] # Age:Sex interaction
```

The census age and sex proportion are inputted into the model to find the predicted mean heart rate.

```
# Compute expected HR using weighted means
expected_HR <- beta_0 + beta_1 * (mean(Masindi_census$Age)) +
  beta_2 * proportion_male + beta_3 * (proportion_male * mean_age_male)

# Print result
expected_HR # 85.4588
```

```
## (Intercept)
##      85.4588
```

```
mean(Masindi$hr) # 85.85728
```

```
## [1] 85.85728
```

The predicted mean heart rate for the population (85.4588 bpm) was within 0.4 bpm of the actual mean heart rate for the cohort (85.85728 bpm). So, although the two groups have significantly different demographic structures, this difference alone would not be expected to shift mean heart rates by any medically meaningful magnitude.

NHANES_Masindi_Comparison.R

Various tests were conducted to compare the demographic structure and heart rates between two sources: Masindi (Uganda) and NHANES (US). Age differences were assessed using the Mann-Whitney-Wilcoxon test, while sex differences were evaluated with the chi-square test for independence. Heart rate comparisons were performed using an ANCOVA model, which adjusted for age, sex, and their interaction. The model's assumptions were rigorously tested, and its validity was confirmed through bootstrapping. Additionally, estimated marginal means were calculated and used for post hoc t-tests. After adjustment, heart rates were significantly higher in the Masindi cohort compared to NHANES. These findings remained consistent when the analysis was limited to routine visits from the Masindi cohort.

All Masindi

```
library(tidyverse)
library(car)
library(emmeans)
library(boot)
library(ggpubr)
library(patchwork)
```

```
folder_path <- as.character(here::here()) #>>>THIS SHOULD POINT TO THE 'MAIN' FOLDER<<<#
# Data Import
NHANES <- read_csv(paste0(folder_path, "/Processed Data/NHANES/HR_DEMO_NHANES.csv"))
```

```
## Rows: 26031 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (2): sex, source
## dbl (3): NHANES_ID, hr, age
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
Masindi <- read_csv(paste0(folder_path, "/Processed Data/Masindi/Masindi_data.csv"))
```

```
## Rows: 1059 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (3): sex, reason, source
## dbl (3): Masindi_ID, hr, age
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
Masindi_routineonly <- read_csv(paste0(folder_path, "/Processed Data/Masindi//Masindi_routine_data.csv"))
```

```
## Rows: 388 Columns: 5
## -- Column specification -----
## Delimiter: ","
```

```
## chr (2): sex, source
## dbl (3): Masindi_ID, hr, age
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

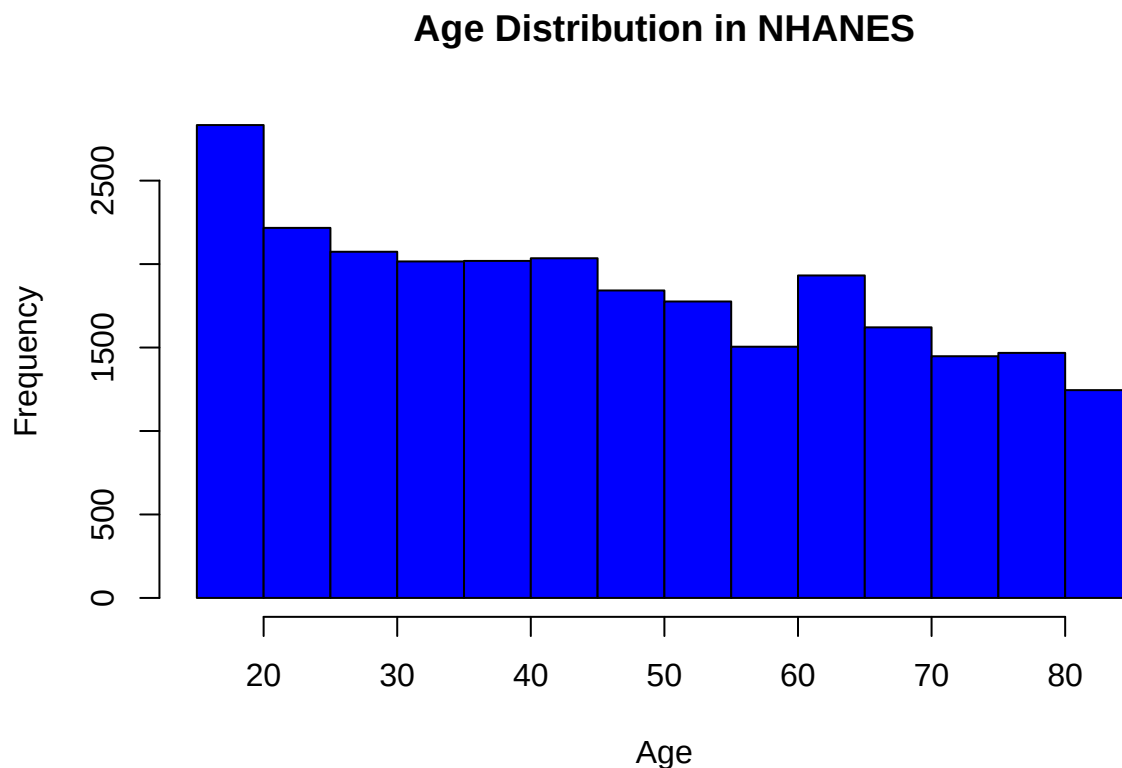
Does the demographic structure (age and sex) of Masindi study participants differ from that of the NHANES study?

```
#### Does the demographic structure (age and sex) of
#### Masindi study participants differ from that of the
#### NHANES study?####

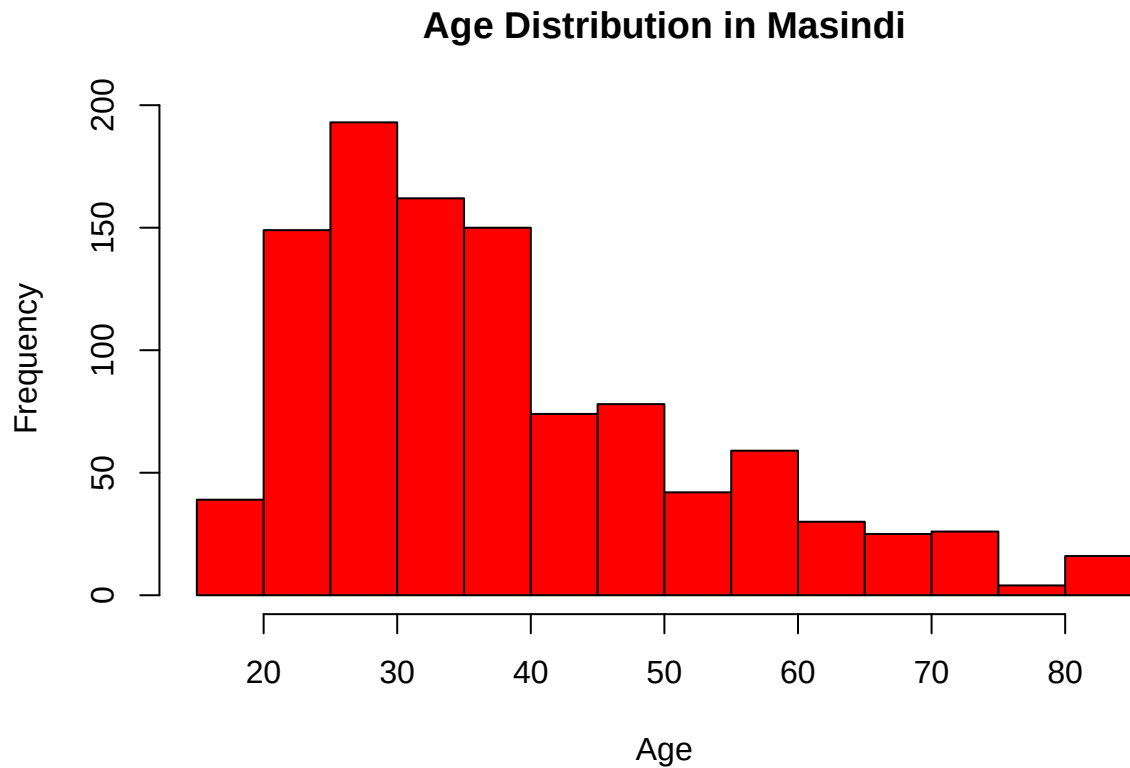
# Does Age differ? - Mann-Whitney-Wilcoxon test
# (non-parametric t-test)

Masindi <- Masindi %>%
  filter(age <= 85) # To match NHANES age range

hist(NHANES$age, main = "Age Distribution in NHANES", xlab = "Age",
  col = "blue") #Non-normal
```



```
hist(Masindi$age, main = "Age Distribution in Masindi", xlab = "Age",
  col = "red") #Non-normal
```



```
wilcox.test(NHANES$age, Masindi$age) # p < .01, reject null, significant difference in ages
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: NHANES$age and Masindi$age
## W = 16626649, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

Significant difference ($p < .01$) in ages between Masindi and NHANES.

```
#Does sex differ? - Chi-square test for independence

table_sex <- table( # Contingency table
  factor(c(NHANES$sex, Masindi$sex), levels = c("M", "F")),
  factor(c(rep("NHANES", length(NHANES$sex)), rep("Masindi", length(Masindi$sex))), levels = c("NHANES", "Masindi"))
)

chi_square_test <- chisq.test(table_sex)

print(chi_square_test) # p < .01, reject null, significant difference in sex ratios
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
```

```
##
## data: table_sex
## X-squared = 34.66, df = 1, p-value = 3.927e-09
```

Significant difference ($p < .01$) in sex ratios between Masindi and NHANES.

The comparison of demographic structures is visualized using a population pyramid.

```
# Representation with population pyramid
Masindi_pyr <- Masindi %>%
  filter(age <= 85) %>%
  mutate(age_category = cut(age, breaks = c(17, 29, 39, 49,
    59, 69, 85), labels = c("18-29", "30-39", "40-49", "50-59",
    "60-69", "70-85")))

NHANES_pyr <- NHANES %>%
  mutate(age_category = cut(age, breaks = c(17, 29, 39, 49,
    59, 69, 85), labels = c("18-29", "30-39", "40-49", "50-59",
    "60-69", "70-85")))

Masindi_summary_df <- Masindi_pyr %>%
  filter(!is.na(age_category)) %>%
  group_by(sex, age_category) %>%
  summarise(count = n()) %>%
  spread(key = sex, value = count, fill = 0)

NHANES_summary_df <- NHANES_pyr %>%
  filter(!is.na(age_category)) %>%
  group_by(sex, age_category) %>%
  summarise(count = n()) %>%
  spread(key = sex, value = count, fill = 0)

pop_pyramid_masindi <- ggplot(data = Masindi_summary_df, aes(x = age_category)) +
  geom_bar(aes(y = M, fill = "Male"), stat = "identity", position = "identity",
    width = 1) + geom_bar(aes(y = -F, fill = "Female"), stat = "identity",
    position = "identity", width = 1) + scale_fill_manual(values = c(Male = "gold2",
    Female = "red"), name = "Gender") + coord_flip() + scale_y_continuous(breaks = scales::pretty_breaks(),
    expand = expansion(mult = 0.05, add = 0), labels = abs, limits = c(-max(abs(Masindi_summary_df$M),
    abs(Masindi_summary_df$F)), max(abs(Masindi_summary_df$M),
    abs(Masindi_summary_df$F)))) + labs(title = "Masindi Study Cohort",
    y = "Count", x = "Age Category") + theme_pubr() + theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line.y = element_line(color = "black"),
    axis.ticks.y = element_line(color = "black"), axis.title.x.bottom = element_blank(),
    axis.line.x.bottom = element_blank(), axis.title.y = element_text(margin = margin(r = 10)),
    legend.position = c(0.81, 0.8), legend.direction = "horizontal",
    legend.title = element_blank(), plot.title = element_text(hjust = 0.5))

pop_pyramid_NHANES <- ggplot(data = NHANES_summary_df, aes(x = age_category)) +
  geom_bar(aes(y = M, fill = "Male"), stat = "identity", position = "identity",
    width = 1) + geom_bar(aes(y = -F, fill = "Female"), stat = "identity",
    position = "identity", width = 1) + scale_fill_manual(values = c(Male = "gold2",
    Female = "red"), name = "Gender") + coord_flip() + scale_y_continuous(breaks = scales::pretty_breaks(),
    expand = expansion(mult = 0.05, add = 0), labels = abs, limits = c(-max(abs(NHANES_summary_df$M),
```

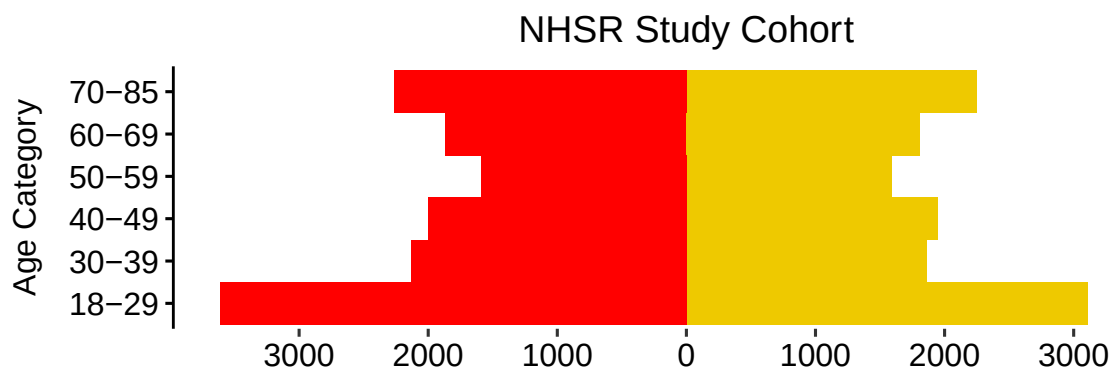
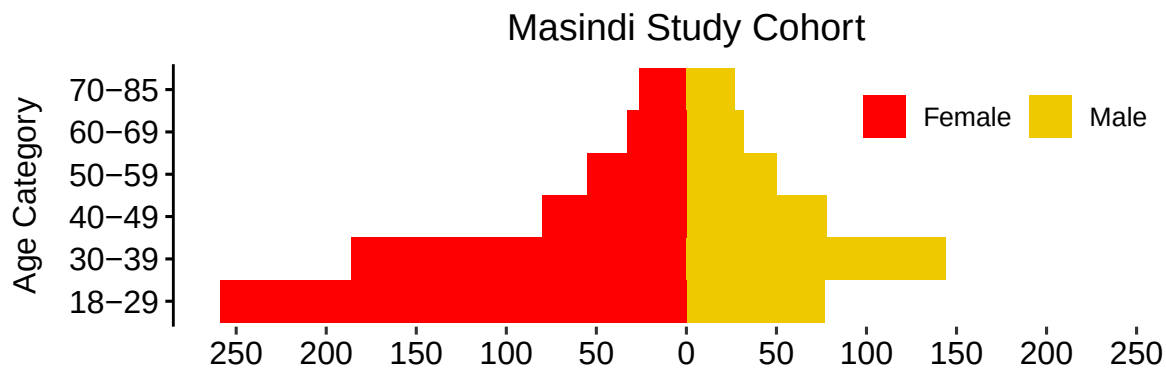
```

    abs(NHANES_summary_df$F)), max(abs(NHANES_summary_df$M),
    abs(NHANES_summary_df$F))) + labs(title = "NHSR Study Cohort",
y = "Count", x = "Age Category") + theme_pubr() + theme(panel.grid.major = element_blank(),
panel.grid.minor = element_blank(), axis.line.y = element_line(color = "black"),
axis.ticks.y = element_line(color = "black"), axis.title.x.bottom = element_blank(),
axis.line.x.bottom = element_blank(), axis.title.y = element_text(margin = margin(r = 10)),
legend.position = "none", legend.title = element_blank(),
plot.title = element_text(hjust = 0.5))

aligned_plots <- pop_pyramid_masindi + plot_spacer() + pop_pyramid_NHANES +
  plot_layout(heights = c(1, 0.1, 1), ncol = 1)

# tiff('FigD.tiff', units='px', width=2200, height=2000,
# res=400)
aligned_plots

```



```
# dev.off()
```

Does heart rate differ between the two groups after adjustment for sex and age?

```

#### Does heart rate differ between the two groups (after
#### adjustment for demographics)?####
Masindi <- Masindi[-5]
Combined_Df <- rbind(NHANES[-1], Masindi[-1])
Combined_Df$sex <- as.factor(Combined_Df$sex)

```

```

Combined_Df$source <- as.factor(Combined_Df$source)

# ANCOVA
ancova_model <- aov(hr ~ source + age * sex, data = Combined_Df)
summary(ancova_model) # All p < .01, reject null, significant difference in heart rates between sources

##              Df  Sum Sq Mean Sq F value Pr(>F)
## source         1  168748  168748   1095.2 <2e-16 ***
## age            1  103406  103406    671.1 <2e-16 ***
## sex            1  145547  145547    944.6 <2e-16 ***
## age:sex        1   34206   34206    222.0 <2e-16 ***
## Residuals     27073 4171454    154
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

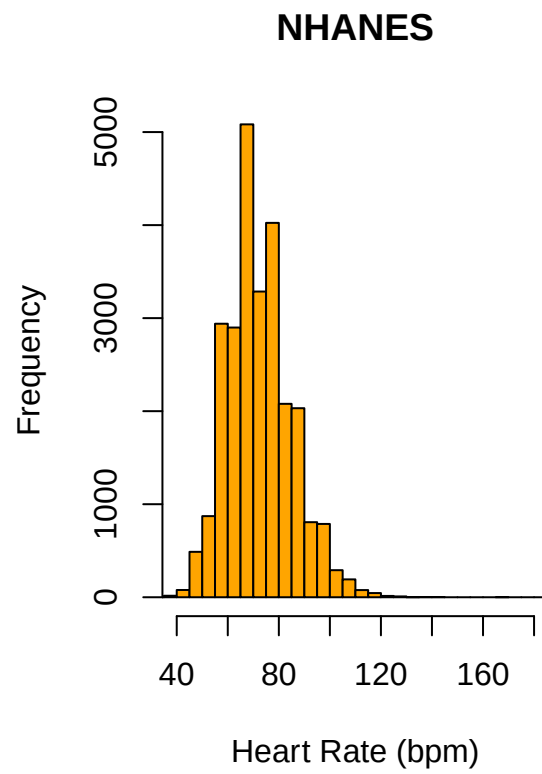
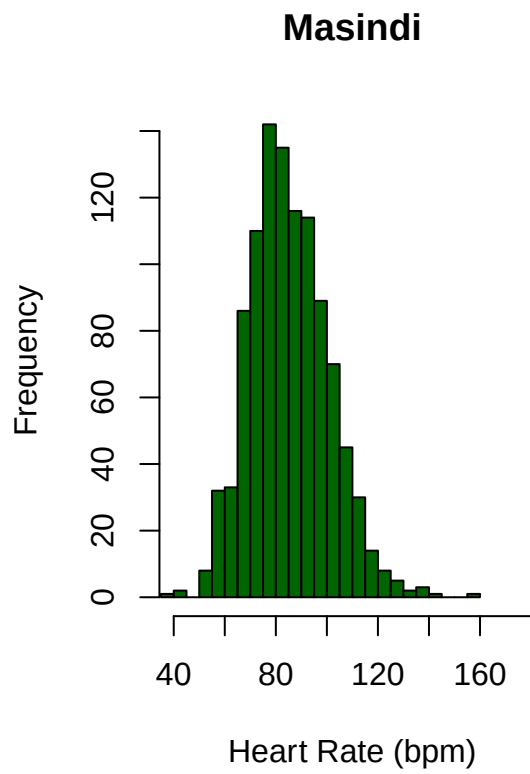
Because there is a continuous outcome variable ('hr') and both categorical ('source' and 'sex') and continuous ('age') predictors, an analysis of covariance (ANCOVA) model was used to test for differences in heart rate between the two groups. All predictor variables were found to be significant ($p < .01$), indicating a significant difference in heart rates between Masindi and NHANES cohorts.

Numerous assumptions were evaluated to ensure the validity of the ANCOVA model.

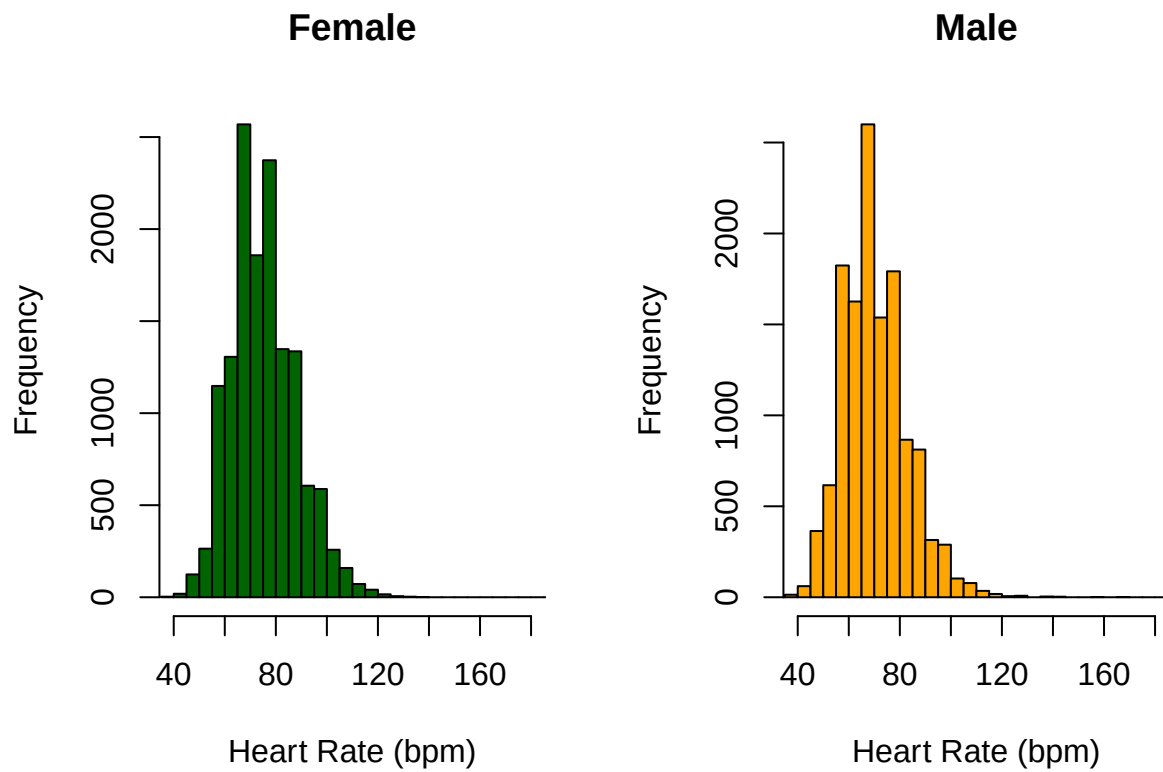
```

#Testing Assumptions
# 1. Normally Distributed Outcome Variable
par(mfrow = c(1, 2))
hist(Combined_Df$hr[Combined_Df$source == "Masindi"], main = "Masindi", xlab = "Heart Rate (bpm)", col = "red")
hist(Combined_Df$hr[Combined_Df$source == "NHANES"], main = "NHANES", xlab = "Heart Rate (bpm)", col = "blue")

```

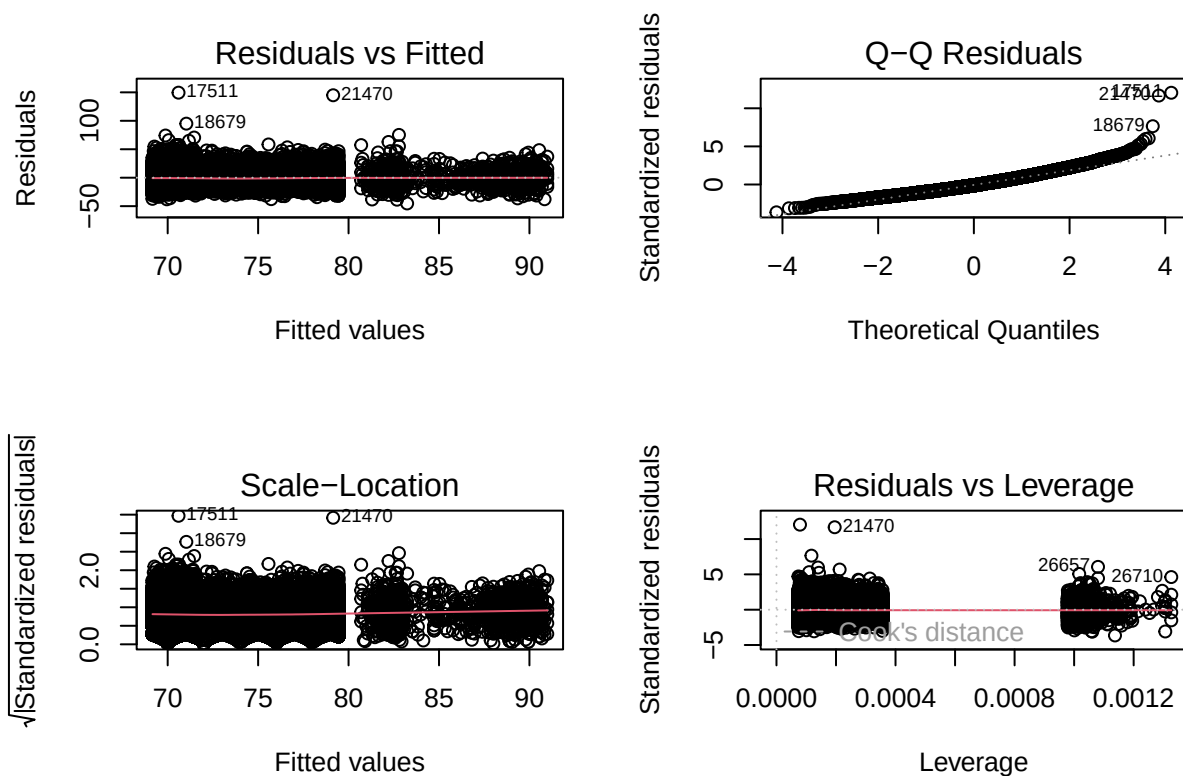


```
hist(Combined_Df$hr[Combined_Df$sex == "F"], main = "Female", xlab = "Heart Rate (bpm)", col = "darkgreen",  
hist(Combined_Df$hr[Combined_Df$sex == "M"], main = "Male", xlab = "Heart Rate (bpm)", col = "orange",
```

```
# HRs are approx. normally distributed across groups

# 2. Normality of Residuals
par(mfrow = c(2, 2))
plot(ancova_model) # Q-Q plot suggests normality of residuals assumption is met, also few outliers
```

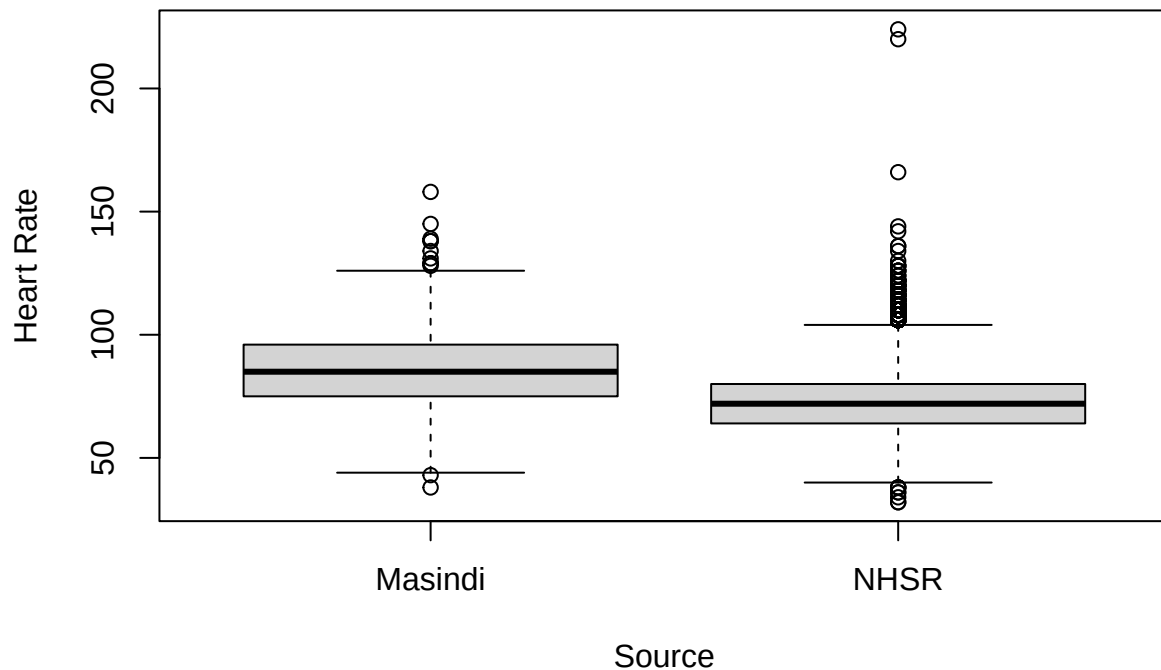


```
par(mfrow = c(1, 1))

# 3. No Multicollinearity
vif(ancova_model, type = "predictor") # GVIF values close to 1 -> No Multicollinearity
```

```
##          GVIF Df GVIF^(1/(2*Df)) Interacts With Other Predictors
## source 1.008617  1      1.004299      --          age, sex
## age    1.008617  3      1.001431      sex          source
## sex    1.008617  3      1.001431      age          source
```

```
# 4. Homogeneity of Variances
#tiff("FigE.tiff", units="px", width=2100, height=2100, res=400)
boxplot(hr ~ source, xlab = "Source", ylab = "Heart Rate", data = Combined_Df, names = c("Masindi", "NH"))
```



```
#dev.off()
bartlett.test(hr ~ source, data = Combined_Df) # Test of homoscedasticity, p < .01 indicates difference
```

```
##
## Bartlett test of homogeneity of variances
##
## data: hr by source
## Bartlett's K-squared = 87.392, df = 1, p-value < 2.2e-16
```

```
leveneTest(hr ~ source, data = Combined_Df) # Robust confirmation of heteroscedasticity, p < .01
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  1  84.128 < 2.2e-16 ***
##      27076
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
var_group <- aggregate(hr ~ source, data = Combined_Df, var)
var_ratio <- max(var_group$hr) / min(var_group$hr)
print(var_ratio) # Variance ratio 1.48 < 1.5, not a severe difference
```

```
## [1] 1.480043
```

```
# Confirming results with bootstrapping
boot_ancova_p <- function(data, indices) {
  boot_sample <- data[indices, ] # Resampling
  model <- aov(hr ~ source + age * sex, data = boot_sample)
  summary_aov <- summary(model)
  p_value <- summary_aov[[1]]$`Pr(>F)`[1] # p-value for `source`

  return(p_value)
}

# Perform bootstrapping for p-values
set.seed(123)
boot_results_p <- boot(data = Combined_Df, statistic = boot_ancova_p, R = 1000)
mean(boot_results_p$t <= 0.05) # Proportion of p-values <= .05 is 1; all p-values are significant
```

```
## [1] 1
```

```
# Bootstrapping indicates results are robust to heteroscedasticity

# 5. Linear Relationship with Covariate (Age)
NHANES_M <- Combined_Df[Combined_Df$source == "NHANES" & Combined_Df$sex == "M", ]
NHANES_F <- Combined_Df[Combined_Df$source == "NHANES" & Combined_Df$sex == "F", ]
Masindi_M <- Combined_Df[Combined_Df$source == "Masindi" & Combined_Df$sex == "M", ]
Masindi_F <- Combined_Df[Combined_Df$source == "Masindi" & Combined_Df$sex == "F", ]

NHANES_M_lr <- lm(hr ~ age, data = NHANES_M)
NHANES_F_lr <- lm(hr ~ age, data = NHANES_F)
Masindi_M_lr <- lm(hr ~ age, data = Masindi_M)
Masindi_F_lr <- lm(hr ~ age, data = Masindi_F)

summary(NHANES_M_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = NHANES_M)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.938  -8.787  -1.270   7.361 149.398
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  72.26343    0.28578  252.869  < 2e-16 ***
## age         -0.03692    0.00556   -6.639 3.29e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.47 on 12568 degrees of freedom
## Multiple R-squared:  0.003495, Adjusted R-squared:  0.003415
## F-statistic: 44.08 on 1 and 12568 DF, p-value: 3.29e-11
```

```
summary(NHANES_F_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = NHANES_F)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.719  -8.536  -1.205   7.308 144.881
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 82.075207   0.262823   312.28  <2e-16 ***
## age        -0.147829   0.005202   -28.42  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.13 on 13459 degrees of freedom
## Multiple R-squared:  0.05661,    Adjusted R-squared:  0.05654
## F-statistic: 807.6 on 1 and 13459 DF,  p-value: < 2.2e-16
```

```
summary(Masindi_M_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_M)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -38.065 -11.141  -2.045   9.833  75.975
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 81.66781    2.33772   34.935  <2e-16 ***
## age         0.01322    0.05247    0.252    0.801
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.79 on 406 degrees of freedom
## Multiple R-squared:  0.0001564,    Adjusted R-squared:  -0.002306
## F-statistic: 0.06352 on 1 and 406 DF,  p-value: 0.8012
```

```
summary(Masindi_F_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_F)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -43.651  -9.965  -1.093   9.657  47.705
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 95.42817    1.58776  60.102  < 2e-16 ***
## age         -0.19681    0.04054  -4.855 1.52e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.55 on 637 degrees of freedom
## Multiple R-squared:  0.03568,    Adjusted R-squared:  0.03416
## F-statistic: 23.57 on 1 and 637 DF,  p-value: 1.519e-06
```

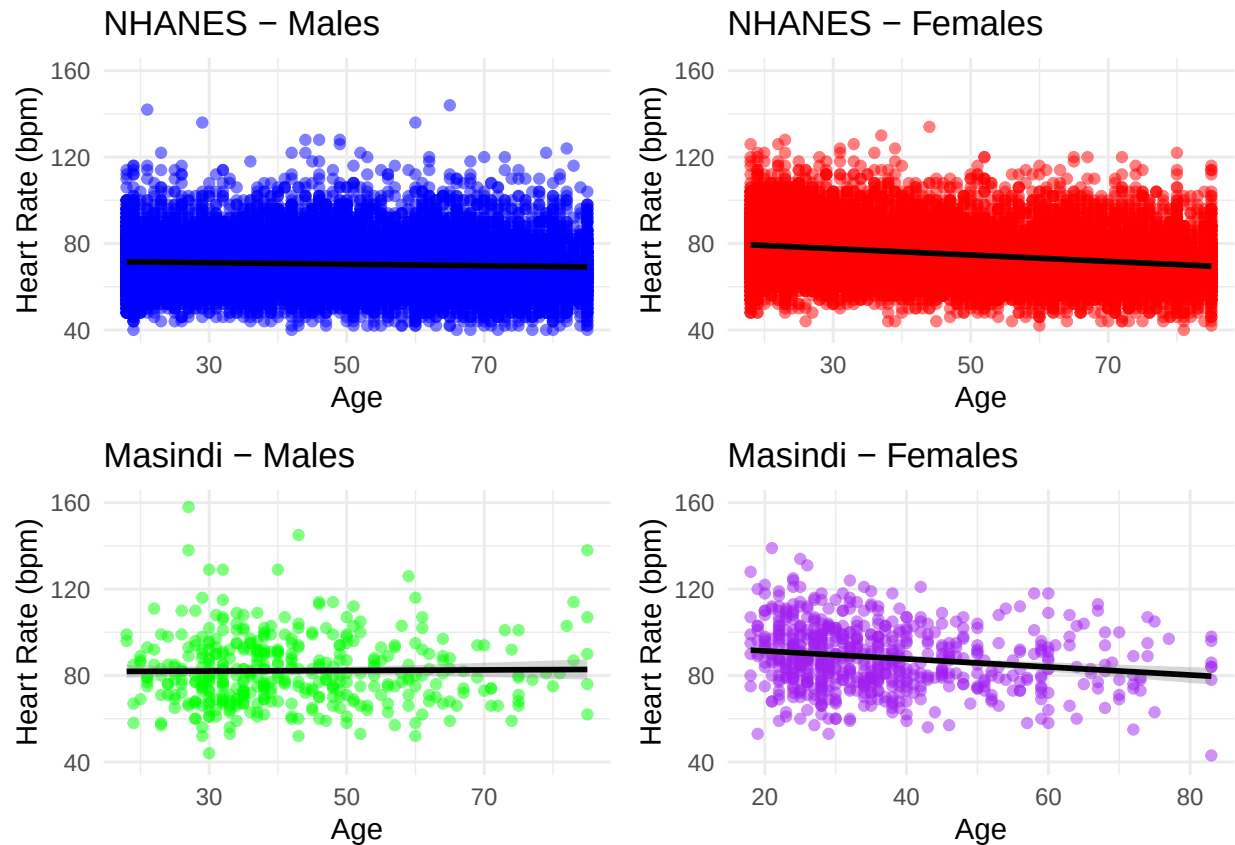
```
p1 <- ggplot(NHANES_M, aes(x = age, y = hr)) +
  geom_point(color = "blue", alpha = 0.5) + # Scatterplot points
  geom_smooth(method = "lm", color = "black", se = TRUE) + # Regression line
  labs(title = "NHANES - Males", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p2 <- ggplot(NHANES_F, aes(x = age, y = hr)) +
  geom_point(color = "red", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "NHANES - Females", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p3 <- ggplot(Masindi_M, aes(x = age, y = hr)) +
  geom_point(color = "green", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "Masindi - Males", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p4 <- ggplot(Masindi_F, aes(x = age, y = hr)) +
  geom_point(color = "purple", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "Masindi - Females", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

gridExtra::grid.arrange(p1, p2, p3, p4, nrow = 2, ncol = 2)
```



```
# All linear with significant p-values and similar variance across ages

# 6. Homogeneity of regression slopes
interaction_model <- lm(hr ~ source * age + sex * age, data = Combined_Df)
reduced_model <- lm(hr ~ source + sex * age, data = Combined_Df)
anova(reduced_model, interaction_model) # p > .05, relationship between age and hr does not differ by s
```

```
## Analysis of Variance Table
##
## Model 1: hr ~ source + sex * age
## Model 2: hr ~ source * age + sex * age
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1  27073 4171454
## 2  27072 4171440   1    13.775 0.0894 0.7649
```

The assumptions of the ANCOVA model are met, indicating that the model was appropriate for the data. Robustness of ANCOVA results were confirmed with bootstrapping: 1000/1000 re-samplings resulted in $p \leq 0.05$.

```
# Post-Hoc: Compute estimated marginal means for 'source',
# adjusting for covariates
emms <- emmeans(ancova_model, ~source, cov.reduce = mean)
summary(emms)
```

```
##   source emmean    SE    df lower.CL upper.CL
```

```
## Masindi 84.41 0.385 27073 83.65 85.17
## NHANES 72.87 0.077 27073 72.72 73.02
##
## Results are averaged over the levels of: sex
## Confidence level used: 0.95
```

```
pairs(emms)
```

```
## contrast      estimate      SE    df t.ratio p.value
## Masindi - NHANES      11.5 0.393 27073  29.364  <.0001
##
## Results are averaged over the levels of: sex
```

The estimated marginal means for heart rate were significantly different between the Masindi and NHANES cohorts ($p < .01$). The mean adjusted heart rate for Masindi was 84.41 bpm with a 95% CI of [83.65, 85.17], while the mean adjusted heart rate for NHANES was 72.87 bpm with a 95% CI of [72.72, 73.02], a difference of 11.54 bpm.

Routine Visits Only

```
#### After removing 'new concern' visits, do heart rates
#### differ between the two groups?####
Masindi_routineonly <- Masindi_routineonly %>%
  filter(age <= 85)

Combined_Df_routine <- rbind(NHANES[-1], Masindi_routineonly[-1])
Combined_Df_routine$sex <- as.factor(Combined_Df_routine$sex)
Combined_Df_routine$source <- as.factor(Combined_Df_routine$source)

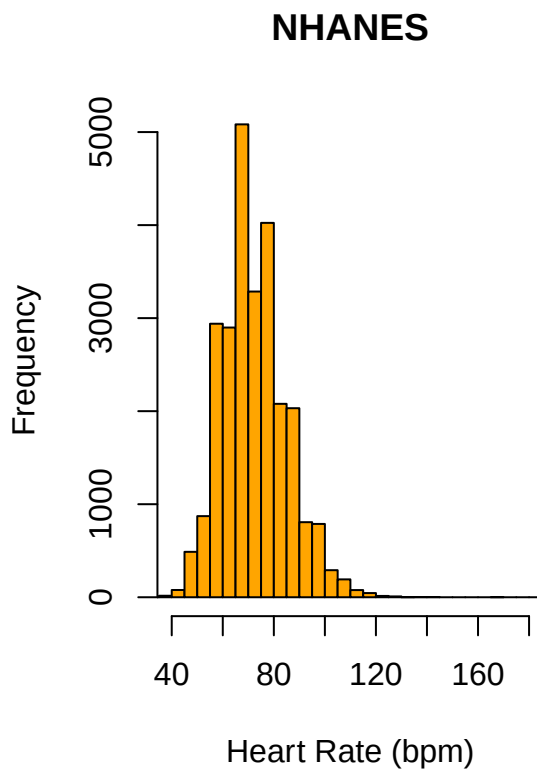
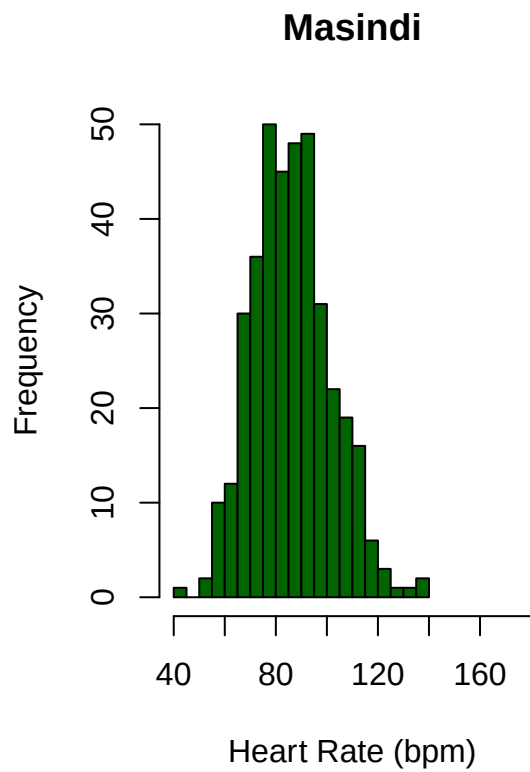
# ANCOVA
ancova_model_routine <- aov(hr ~ source + age * sex, data = Combined_Df_routine)
summary(ancova_model_routine) # All p < .01, reject null, significant difference in heart rates between
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## source         1   73793   73793    485.0 <2e-16 ***
## age            1  101672  101672    668.2 <2e-16 ***
## sex            1  141265  141265    928.4 <2e-16 ***
## age:sex        1   33336   33336    219.1 <2e-16 ***
## Residuals    26410 4018347     152
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

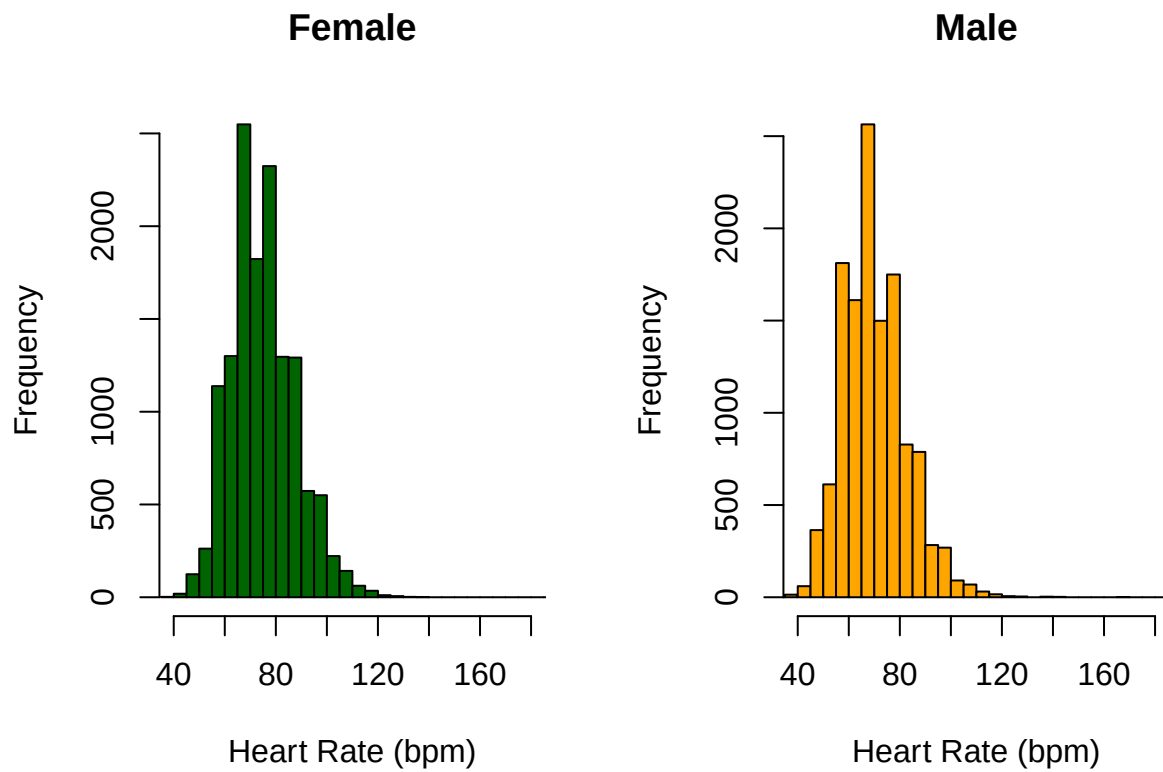
Significant difference ($p < .01$) in heart rates between Masindi and NHANES cohorts after excluding “new concern” visits.

```
#Testing Assumptions

# 1. Normally Distributed Outcome Variable
par(mfrow = c(1, 2))
hist(Combined_Df_routine$hr[Combined_Df_routine$source == "Masindi"], main = "Masindi", xlab = "Heart Rate")
hist(Combined_Df_routine$hr[Combined_Df_routine$source == "NHANES"], main = "NHANES", xlab = "Heart Rate")
```

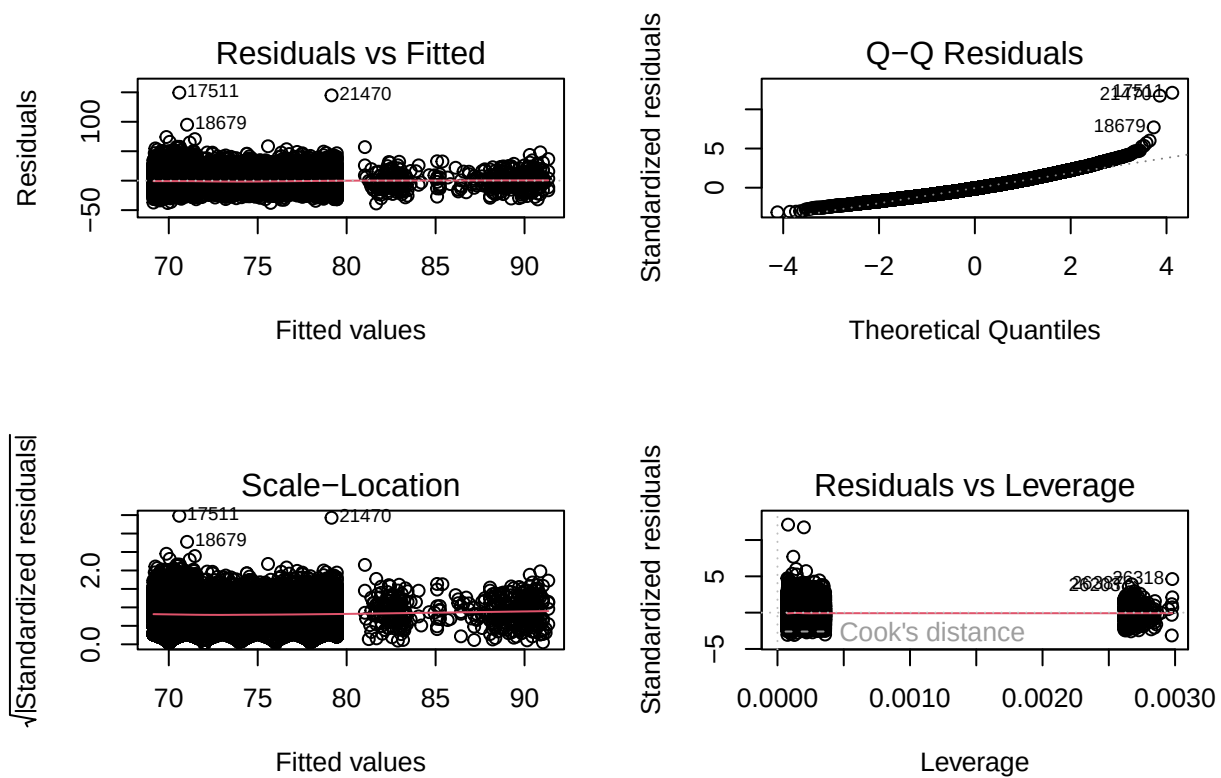



```
hist(Combined_Df_routine$hr[Combined_Df_routine$sex == "F"], main = "Female", xlab = "Heart Rate (bpm)", col = "red", border = "black", freq = FALSE)
hist(Combined_Df_routine$hr[Combined_Df_routine$sex == "M"], main = "Male", xlab = "Heart Rate (bpm)", col = "blue", border = "black", freq = FALSE)
```



```
# HRs are approx. normally distributed across groups

# 2. Normality of Residuals
par(mfrow = c(2, 2))
plot(ancova_model_routine) # Q-Q plot suggests normality of residuals assumption is met, also few outliers
```

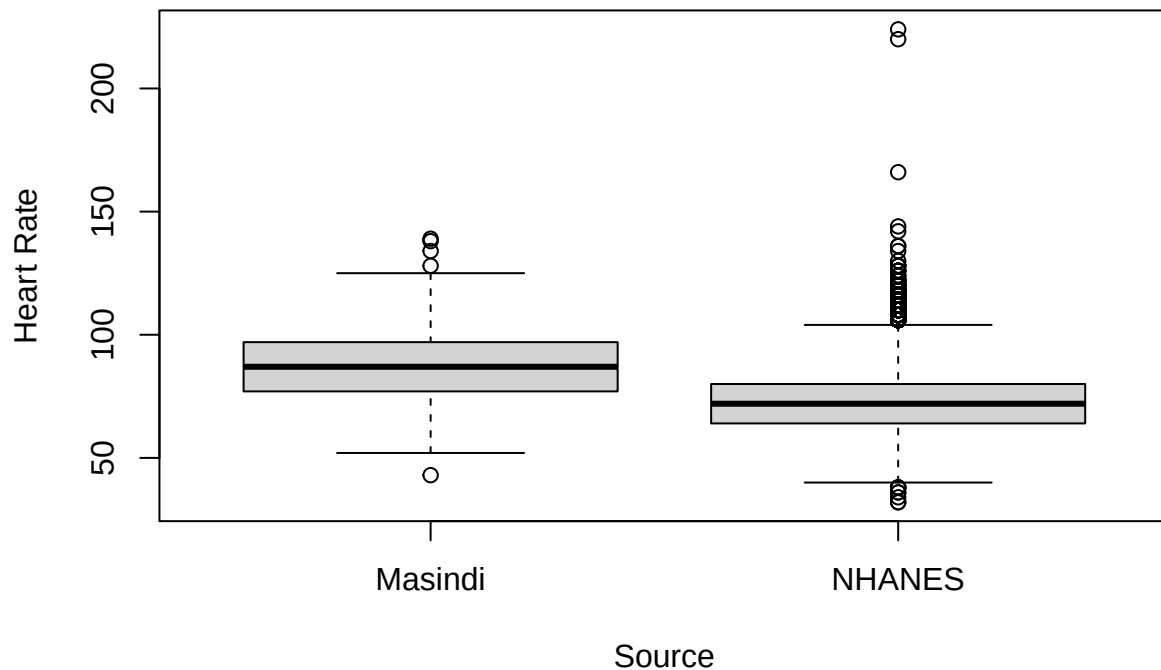


```
par(mfrow = c(1, 1))

# 3. No Multicollinearity
vif(ancova_model_routine, type = "predictor") # GVIF values close to 1 -> No Multicollinearity
```

```
##          GVIF Df GVIF^(1/(2*Df)) Interacts With Other Predictors
## source 1.00624  1      1.003115      --          age, sex
## age    1.00624  3      1.001037      sex          source
## sex    1.00624  3      1.001037      age          source
```

```
# 4. Homogeneity of Variances
boxplot(hr ~ source, xlab = "Source", ylab = "Heart Rate", data = Combined_Df_routine) # Approx. homogeneity
```



```
bartlett.test(hr ~ source, data = Combined_Df_routine) # Test of homoscedasticity, p < .01 indicates di
```

```
##
## Bartlett test of homogeneity of variances
##
## data: hr by source
## Bartlett's K-squared = 30.147, df = 1, p-value = 4.005e-08
```

```
leveneTest(hr ~ source, data = Combined_Df_routine) # Robust confirmation of heteroscedasticity, p < .01
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  1 30.478 3.408e-08 ***
##      26413
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
var_group <- aggregate(hr ~ source, data = Combined_Df_routine, var)
var_ratio <- max(var_group$hr) / min(var_group$hr)
print(var_ratio) # Variance ratio 1.46 < 1.5, not a severe difference
```

```
## [1] 1.455918
```

```

# Confirming results with bootstrapping
boot_ancova_p_adj <- function(data, indices) { #Adjusted for sample sizes
  nhanes_data <- data[data$source == "NHANES", ]
  masindi_data <- data[data$source == "Masindi", ]
  nhanes_indices <- sample(seq_len(nrow(nhanes_data)), size = 700, replace = TRUE) #Adjusted for large
  masindi_indices <- sample(seq_len(nrow(masindi_data)), size = 300, replace = TRUE)
  boot_sample <- rbind(nhanes_data[nhanes_indices, ], masindi_data[masindi_indices, ])
  model <- aov(hr ~ source + age * sex, data = boot_sample)
  summary_aov <- summary(model)
  p_value <- summary_aov[[1]]$`Pr(>F)`[1]

  return(p_value)
}

# Perform bootstrapping for p-values
boot_results_p_adj <- boot(data = Combined_Df_routine, statistic = boot_ancova_p_adj, R = 1000)
mean(boot_results_p_adj$t <= 0.05) # Proportion of p-values <= 0.05 is 1; all p-values are significant

```

```
## [1] 1
```

```

#Bootstrapping indicates results are robust to heteroscedasticity

# 5. Linear Relationship with Covariate (Age)
NHANES_M_routine <- Combined_Df_routine[Combined_Df_routine$source == "NHANES" & Combined_Df_routine$se
NHANES_F_routine <- Combined_Df_routine[Combined_Df_routine$source == "NHANES" & Combined_Df_routine$se
Masindi_M_routine <- Combined_Df_routine[Combined_Df_routine$source == "Masindi" & Combined_Df_routine$
Masindi_F_routine <- Combined_Df_routine[Combined_Df_routine$source == "Masindi" & Combined_Df_routine$

NHANES_M_routine_lr <- lm(hr ~ age, data = NHANES_M_routine)
NHANES_F_routine_lr <- lm(hr ~ age, data = NHANES_F_routine)
Masindi_M_routine_lr <- lm(hr ~ age, data = Masindi_M_routine)
Masindi_F_routine_lr <- lm(hr ~ age, data = Masindi_F_routine)

summary(NHANES_M_routine_lr) # p < .01

```

```

##
## Call:
## lm(formula = hr ~ age, data = NHANES_M_routine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.938  -8.787  -1.270   7.361 149.398
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 72.26343    0.28578 252.869  < 2e-16 ***
## age        -0.03692    0.00556  -6.639 3.29e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.47 on 12568 degrees of freedom
## Multiple R-squared:  0.003495, Adjusted R-squared:  0.003415
## F-statistic: 44.08 on 1 and 12568 DF, p-value: 3.29e-11

```

```
summary(NHANES_F_routine_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = NHANES_F_routine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37.719  -8.536  -1.205   7.308 144.881
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 82.075207   0.262823   312.28  <2e-16 ***
## age        -0.147829   0.005202   -28.42  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.13 on 13459 degrees of freedom
## Multiple R-squared:  0.05661,    Adjusted R-squared:  0.05654
## F-statistic: 807.6 on 1 and 13459 DF,  p-value: < 2.2e-16
```

```
summary(Masindi_M_routine_lr) # p > .05, NOT significant
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_M_routine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.312 -11.830  -1.213   8.719  54.462
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 79.36710    4.05832   19.557  <2e-16 ***
## age          0.04907    0.08326    0.589    0.557
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.74 on 109 degrees of freedom
## Multiple R-squared:  0.003177,    Adjusted R-squared: -0.005968
## F-statistic: 0.3474 on 1 and 109 DF,  p-value: 0.5568
```

```
summary(Masindi_F_routine_lr) # p < .01
```

```
##
## Call:
## lm(formula = hr ~ age, data = Masindi_F_routine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -35.559  -9.182  -1.174   9.039  46.719
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 96.92806    2.36890  40.917  < 2e-16 ***
## age         -0.22131    0.06178  -3.582 0.000404 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.77 on 271 degrees of freedom
## Multiple R-squared:  0.04521,    Adjusted R-squared:  0.04169
## F-statistic: 12.83 on 1 and 271 DF,  p-value: 0.0004037
```

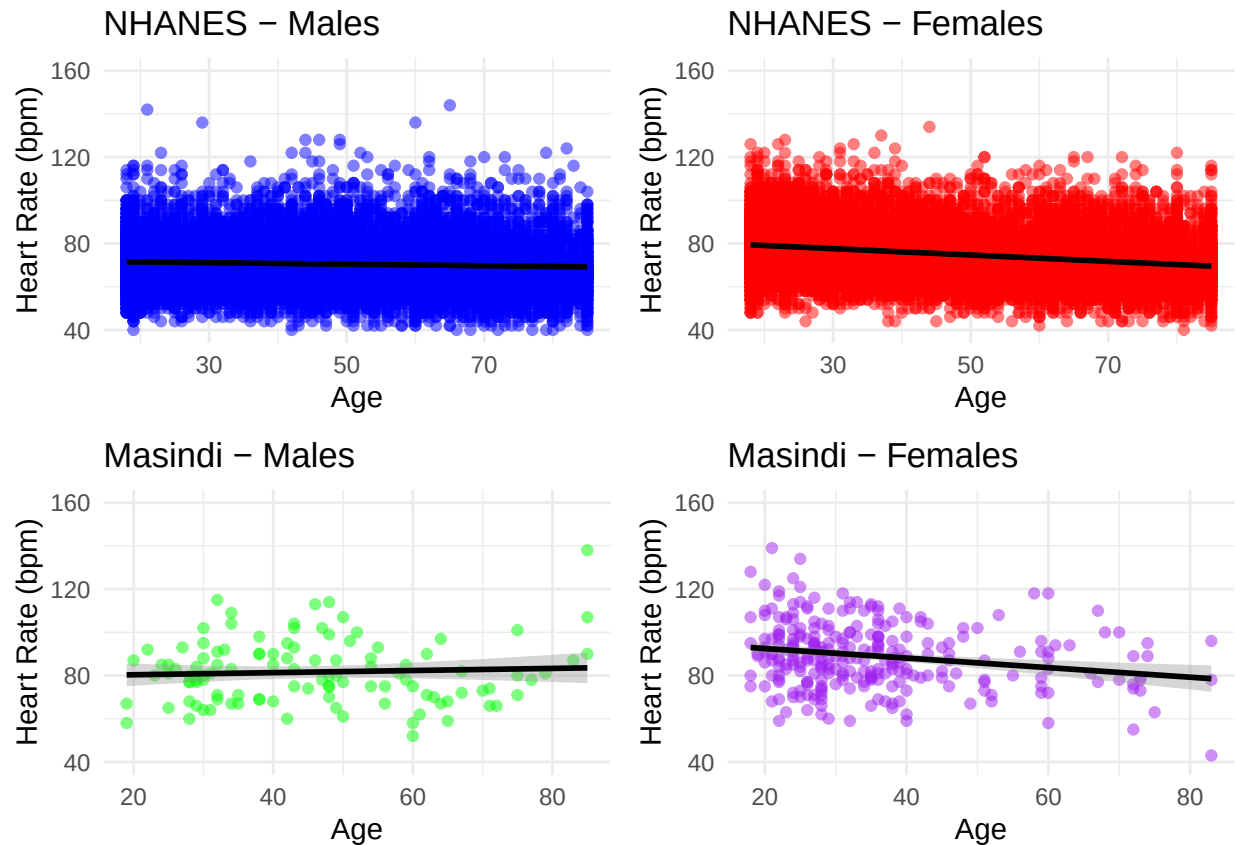
```
p1 <- ggplot(NHANES_M_routine, aes(x = age, y = hr)) +
  geom_point(color = "blue", alpha = 0.5) + # Scatterplot points
  geom_smooth(method = "lm", color = "black", se = TRUE) + # Regression line
  labs(title = "NHANES - Males", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p2 <- ggplot(NHANES_F_routine, aes(x = age, y = hr)) +
  geom_point(color = "red", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "NHANES - Females", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p3 <- ggplot(Masindi_M_routine, aes(x = age, y = hr)) +
  geom_point(color = "green", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "Masindi - Males", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

p4 <- ggplot(Masindi_F_routine, aes(x = age, y = hr)) +
  geom_point(color = "purple", alpha = 0.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "Masindi - Females", x = "Age", y = "Heart Rate (bpm)") +
  theme_minimal() +
  ylim(40, 160)

gridExtra::grid.arrange(p1, p2, p3, p4, nrow = 2, ncol = 2)
```



```
# Slope = 0 for Males in Masindi, while positive slope for Females in Masindi. Accounted for in interaction
# 6. Homogeneity of regression slopes
interaction_routine_model <- lm(hr ~ source * age + sex * age, data = Combined_Df_routine)
reduced_routine_model <- lm(hr ~ source + sex * age, data = Combined_Df_routine)
anova(reduced_routine_model, interaction_routine_model) # p > .05, relationship between age and hr does
```

```
## Analysis of Variance Table
##
## Model 1: hr ~ source + sex * age
## Model 2: hr ~ source * age + sex * age
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1  26410 4018347
## 2  26409 4018285   1    61.401 0.4035 0.5253
```

As before, the assumptions of the ANCOVA model are met, indicating that the model was appropriate for the (reduced) data. Robustness of ANCOVA results were confirmed with bootstrapping: 1000/1000 re-samplings resulted in $p \leq 0.05$.

```
# Post-Hoc: Compute estimated marginal means for 'source',
# adjusting for covariates
emms <- emmeans(ancova_model_routine, ~source, cov.reduce = mean)
summary(emms)
```

```
##   source emmean      SE    df lower.CL upper.CL
```



```
## Masindi 84.74 0.6320 26410 83.5 85.98
## NHANES 72.85 0.0765 26410 72.7 73.00
##
## Results are averaged over the levels of: sex
## Confidence level used: 0.95
```

```
pairs(emms)
```

```
## contrast estimate SE df t.ratio p.value
## Masindi - NHANES 11.9 0.636 26410 18.686 <.0001
##
## Results are averaged over the levels of: sex
```

After removal of "new concern" cases, the estimated marginal means for heart rate still significantly differed between Masindi and NHANES cohorts ($p < .01$). The mean adjusted heart rate for Masindi was 84.74 with 95% CI of [83.5, 85.98], while the mean adjusted heart rate for NHANES was 72.85 with 95% CI of [72.7, 73.00], a difference of 11.89 bpm.

Summary

Summary of Statistical Results			
Variables	Statistical Test	p-value	Results
Masindi_HR_Analysis.R – Complete Dataset			
Masindi\$sex Masindi\$age	Two-tailed Mann-Whitney-Wilcoxon	< .001	There is a significant difference between the ages of males and females.
Masindi\$sex Masindi\$hr	Two-tailed Mann-Whitney-Wilcoxon	< .001	Heart rates differ significantly by sex. Mean female heart rates are 5.98 bpm higher than in males. Mean female heart rate = 88.20 ± 14.80 bpm Mean male heart rate = 82.22 ± 15.89 bpm
Masindi\$reason Masindi\$hr	Two-tailed Welch's t-test	.109 > .05	Reason for visit does not significantly affect heart rate.
Masindi\$age Masindi\$hr	Linear Regression	< .001	Heart rates differ significantly by age. Without adjustment for sex, heart rate decreases by 0.136 bpm per year of age.
Masindi\$sex Masindi\$age (Masindi\$sex * Masindi\$age) Masindi\$hr	ANCOVA Separate Linear Regressions by sex	< .001 Male: .783 > .05 Female: < .001	The interaction between sex and age on hear rate is significant. In males, age does not significantly impact heart rate. In females, age significantly reduces heart rate by 0.198 bpm per year.
Masindi_HR_Analysis.R – Routine-Only Dataset			
Masindi_routine\$sex Masindi_routine\$age	Two-tailed Mann-Whitney-Wilcoxon	< .001	There is a significant difference between the ages of males and females.
Masindi_routine\$sex Masindi_routine\$hr	Two-tailed Mann-Whitney-Wilcoxon	< .001	Heart rates differ significantly by sex. Mean female heart rates are 7.51 bpm higher than in males. Mean female heart rate = 89.07 ± 15.09 bpm Mean male heart rate = 81.56 ± 14.73 bpm
Masindi_routine\$age Masindi_routine\$hr	Linear Regression	< .001	Heart rates differ significantly by age. Without adjustment for sex, heart rate decreases by 0.173 bpm per year of age.
Masindi_routine\$sex Masindi_routine\$age (Masindi_routine\$sex * Masindi_routine\$age) Masindi_routine\$hr	ANCOVA Separate Linear Regressions by sex	.0091 < .01 Male: .646 > .05 Female: < .001	The interaction between sex and age on hear rate is significant. In males, age does not significantly impact heart rate. In females, age significantly reduces heart rate by 0.221 bpm per year.
Masindi_Census_Comparison.R – Complete Dataset			
Masindi_census\$Age Masindi\$Age	Two-tailed Mann-Whitney-Wilcoxon	< .001	There is a significant difference between the ages of the Masindi district census projection and Masindi cohort.
Masindi_census\$sex Masindi\$sex	Chi-square Test for Independence	< .001	There is a significant difference in sex ratios between the Masindi district census projection and Masindi cohort.
MCL_Male\$Age Masindi_Male\$Age MCL_Female\$Age Masindi_Female\$Age	Separate two-tailed Mann-Whitney-Wilcoxon tests by sex	Male: < .001 Female: < .001	For both males and females, the mean age of the Masindi census significantly differs from the mean age of the clinical cohort.

NHANES_Masindi_Comparison.R – Complete Dataset			
NHANES\$age Masindi\$age	Two-tailed Mann-Whitney-Wilcoxon	< .001	There is a significant difference between the ages of NHANES and Masindi cohorts.
NHANES\$sex Masindi\$sex	Chi-square Test for Independence	< .001	There is a significant difference in sex ratios between NHANES and Masindi cohorts.
Combined_Df\$age Combined_Df\$sex (Combined_Df\$age * Combined_Df\$sex) Combined_Df\$source	ANCOVA Post hoc two-tailed t-test on estimated marginal means	< .001 < .001	After adjusting for age, sex, and their interaction, the heart rate of Masindi significantly differs from NHANES. The mean adjusted heart rate for Masindi was 11.54 bpm higher than for NHANES. The mean adjusted heart rate for Masindi was 84.41 bpm with a 95% CI of [83.65, 85.17]. The mean adjusted heart rate for NHANES was 72.87 bpm with a 95% CI of [72.72, 73.02].
NHANES_Masindi_Comparison.R – Routine-Only Dataset			
Combined_Df _routine\$age Combined_Df _routine\$sex (Combined_Df _routine\$age * Combined_Df _routine\$sex) Combined_Df _routine\$source	ANCOVA Post hoc two-tailed t-test on estimated marginal means	< .001 < .001	After adjusting for age, sex, and their interaction, the heart rate of Masindi significantly differs from NHANES. The mean adjusted heart rate for Masindi was 11.54 bpm higher than for NHANES. The mean adjusted heart rate for Masindi was 84.74 bpm with a 95% CI of [83.5, 85.98]. The mean adjusted heart rate for NHANES was 72.85 bpm with a 95% CI of [72.7, 73.00].